Online Supplement to "Pricing and Rarity Design of Blind Boxes"

C. Proofs from Sections 2 and 4

C.1. Proofs from Section 2

We formalize the notion that consumers' perceived beliefs about rarity under blind box are consistent with the actual realized purchase quantities in equilibrium.

Lemma 5 (Belief is Consistent with Outcome). *For any given blind box (p_b, q), the ratio of the expected number of special items collected by consumers in equilibrium to the total expected quantity of items purchased equals the drawing probability q.*

On the surface, it appears that the different stopping conditions exhibited by consumers (e.g., see Lemma 1) would distort the realized purchase quantities in equilibrium. However, as we argue in the proof of the lemma, along every sample path and with each purchase, a fraction q of consumers who have not yet reached their stopping condition receive the special item. As a result, the corresponding rarity, $r = q/((1-q)/n)$, is also maintained in equilibrium.

Proof of Lemma 5. In this proof, we demonstrate that at equilibrium, once all consumers have finished their purchases, the fraction of special items collected by consumers equals the probability of drawing a special item from the blind box, q. In other words, the consumers' beliefs are consistent with the equilibrium state, in expectation.

First, we present an alternative interpretation of consumers purchasing blind boxes as a stochastic process and introduce pertinent notation. For the sake of this proof, we assume that the purchase of blind boxes takes place over rounds, where in each round τ, a single consumer purchases[8] the blind box if its marginal value, v_B, is not lower than the price, p_b. With some probability, this consumer reaches the stopping condition (outlined in Lemma 1) and ceases purchasing, i.e., drops out. With the remaining probability, the consumer moves on to round $\tau + 1$, and the process continues until they meet their stopping condition. Formally, at the beginning of round $\tau = 1$, the consumers purchase the blind box as long as $v_B \geq p_b$. We first rule out a special case. Because of consumer heterogeneity, for a given price, some consumers may not purchase at any round (e.g., when $v_B < p_b$ at round $\tau = 1$). These consumers can be excluded from the analysis, as they do not affect the equilibrium outcome. Suppose that x_τ denotes the probability that the consumer does not reach their stopping condition before round τ with $x_1 = 1$. Further, let m_τ^s and m_τ be

[8] Although the model contains a unit mass of heterogeneous consumers, it suffices to analyze an arbitrary consumer. We show that, for this consumer, the result holds in every round along every possible sample path. Therefore, the result holds in equilibrium despite consumer heterogeneity.

random variables denoting the number of special items and the total number of items collected by the consumer at the beginning of round $\tau > 1$. Since the consumer receives one item in every round before dropping out, one can derive the expected value of m_τ as follows:

$$E[m_\tau] = \sum_{t=1}^{\tau-1} x_t,$$

where $x_t - x_{t+1}$ is the probability that the consumer reaches their stopping condition at the end of round t.

To prove the required lemma, it is sufficient for us to show $E[m_\tau^s] = qE[m_\tau]$ in the limit $\tau \to \infty$ as the consumer reaches their stopping condition almost certainly. In fact, we show a stronger result, namely that for every $\tau > 1$, $E[m_\tau^s] = qE[m_\tau]$ at the beginning of round τ (or end of round $\tau - 1$). We proceed by induction. For the base case, consider $\tau = 2$. Since the consumer always purchases the blind box in the first round, we have $m_2 = 1$. Moreover, $E[m_2^s] = q$ by definition of the blind box, which completes the base case.

Next, suppose that the inductive claim holds up to round τ and consider round $\tau + 1$, where we need to prove that $E[m_{\tau+1}^s] = qE[m_{\tau+1}]$. First, note that $E[m_{\tau+1}] = E[m_\tau] + x_\tau$, where x_τ is the probability that the consumer receives an item in round τ, i.e., has not reached their stopping condition. In a similar vein, we have $E[m_{\tau+1}^s] = E[m_\tau^s] + qx_\tau$. Since x_τ is the probability that the consumer purchases the blind box, qx_τ is the probability of receiving a special item. To prove the inductive hypothesis, we have

$$E[m_{\tau+1}^s] = E[m_\tau^s] + qx_\tau = qE[m_\tau] + qx_\tau = q(E[m_\tau] + x_\tau) = qE[m_{\tau+1}].$$

The second equation above stems from the inductive hypothesis. This completes the proof. □

Proof of Lemma 1. To delineate the stopping conditions, first recall that for any given blind box (p_b, q), the consumer of type (ℓ_r, ℓ_s) will continue to purchase the blind box as long as

$$(1-q)\left(\frac{n-\ell_r-n^L}{n}v^L + \frac{\ell_r-n^H}{n}v^H\right) + q(1-n_s)v_s \geq p_b,$$

where $n^L \in \{0, \ldots, n-\ell_r\}$, $n^H \in \{0, \ldots, \ell_r\}$, and $n_s \in \{0, 1\}$ denote the number of unique high-value regular, low-value regular, and special items currently in the consumer's possession. This can also be interpreted as the consumer's current state based on past purchases.

Define

$$v_B(n^L, n^H, n_s) = (1-q)(\frac{n-\ell_r-n^L}{n}v^L + \frac{\ell_r-n^H}{n}v^H) + q(1-n_s)v_s$$

to be the consumer's marginal value from purchasing the blind box at state (n^L, n^H, n_s). Clearly, $v_B(n^L, n^H, n_s)$ is decreasing in n^H for any given n^L and n_s. The same monotonicity holds for n^L

and n_s. Next, for any given n^L, we formally define the two cutoffs $\bar{n}^H$ and $\underline{n}^H$ as the smallest values in $[0, \ell_r]$ satisfying

$$v_B(n^L, \bar{n}^H, 0) < p_b \leq v_B(n^L, \bar{n}^H - 1, 0) \quad \text{and} \quad v_B(n^L, \underline{n}^H, 1) < p_b \leq v_B(n^L, \underline{n}^H - 1, 1),$$

respectively. Intuitively, $(n^L, \bar{n}^H)$ is the maximum number of unique regular items the consumer can obtain without the special item before the stopping condition is reached. Analogously, $(n^L, \underline{n}^H)$ denotes the maximum number of unique regular items that can be collected along with the special item. Additionally, we also account for the boundary cases. If $p_b \leq v_B(n - \ell_r, \ell_r, 0)$, the consumer would never stop purchasing even after receiving all $(n - \ell_r, \ell_r)$ unique regular item, in which case, we set $\bar{n}^H = \ell_r + 1$ as convention. We also set $\bar{n}^H = 0$ if $p_b > v_B(0, 0, 0)$ as no purchases would occur when the price is too high. Analogously, if $p_b \leq v_B(n - \ell_r, \ell_r, 1)$, we set $\underline{n}^H = n - \ell + 1$ since the consumer would stop purchasing after collecting all $n + 1$ items. We set $\underline{n}^H = 0$ if $p_b > v_B(0, 0, 1)$.

We now go over each of the stopping conditions in the lemma statement:

(i) $n_s = 0, n^H = \bar{n}^H$. Suppose the consumer does not acquire the special item before receiving at least $(n^L, \bar{n}^H)$ unique regular items. Prior to obtaining the $(n^L, \bar{n}^H)$-th unique regular item, the consumer's marginal valuation for the blind box is $v_B(n^L, \bar{n}^H - 1, 0)$. By definition, this term is at least p_b. Moreover, since v_B is decreasing in n^H, this implies that for all $n^H \leq \bar{n}^H - 1$, the consumer would not stop purchasing if they have (n^L, n^H) unique regular items and no special item, i.e., $p_b \leq v_B(n^L, n^H, 0)$ for all $n^H \leq \bar{n}^H - 1$. Consequently, the consumer would stop purchasing upon receiving the $\bar{n}^H$-th regular item since $p_b > v_B(n^L, n^H, 0)$.

(ii) $n_s = 1, n^H = \underline{n}^H$. In a similar vein, suppose that the consumer has already obtained the special item before purchasing the $\underline{n}^H$-th unique regular item. Once again, by definition of $\underline{n}^H$, and since $v_B(n^L, n^H, 1)$ is decreasing in n^H, it is clear that the consumer would not stop when $n^H \leq \underline{n}^H - 1$. Subsequently, since $p_b > v_B(n^L, \underline{n}^H, 1)$, this state, i.e., $(n^L, \underline{n}^H, 1)$, serves as a stopping point.

(iii) $n_s = 1$ and $n^H \in [\underline{n}^H, \bar{n}^H]$. Suppose that the consumer obtains the special item between receiving the $\underline{n}^H$-th and $\bar{n}^H$-th unique regular item. In this case, the consumer would not stop after receiving the $\underline{n}^H$-th unique regular item (or prior to this) as $p_b \leq v_B(n^L, \underline{n}^H, 0)$ by definition. Further, define $n_H^+ \in [\underline{n}^H, \bar{n}^H - 1]$ to be the number of regular items possessed by the consumer when the special item is drawn for the first time. In this case, we know that $v_B(n^L, n_H^+, 1) < p_b \leq v_B(n^L, n_H^+, 0)$. Therefore, the consumer stops purchasing after drawing the special item for the first time. □

Proof of Lemma 2. Our goal is to prove that separate selling is more profitable than blind box when the number of regular items n approaches infinity, i.e., $\lim_{n\to\infty} \Pi_B^*/\Pi_S^* < 1$. Towards this goal, we first define a proxy solution with profit Π_S^D and show that $\lim_{n\to\infty} \Pi_B^*/\Pi_S^D \le 1/e$. Second, we show that for all $n \ge 1$, the profit under the proxy solution is at most a factor of two larger than the optimal profit under separate selling, i.e., $\Pi_S^D/\Pi_S^* \le 2$, which gives us the desired result.

Step 1. We begin by deriving the optimal profit under blind box as $n \to \infty$, based on Lemma E3. Suppose that $\Pi_B^{-s}(n)$ denotes the optimal blind box profit in the absence of the special item. Then, from Lemma E3, we have

$$\lim_{n\to\infty} \frac{\Pi_B^{-s}(n)}{n} \le \begin{cases} \frac{1}{e}\left(\omega(v^H - c) + (1-\omega)(v^L - c)\right) & \text{if } 0 \le c \le v^L, \\ \frac{\omega(v^H - c)}{e} & \text{if } v^L < c \le v^H. \end{cases}$$

We now show that the same upper bound applies to our setting with the special item. First, note that as $n \to \infty$, the probability q of drawing the special item approaches zero. Therefore, any price $p_b \le q v_s(q, v^H) = q(1 + \alpha(1-r))v^H$ is suboptimal as the total profit does not scale with n.

Next, given any price p_b for the blind box, suppose that in the absence of the special item, the consumer stops purchasing after obtaining n^L, n^H unique low-value and high-value regular items, respectively. As long as $p_b > q v_s(q, v^H)$, it is not difficult to see that even in the presence of the special item, the consumer stops purchasing the blind box after obtaining n^L unique low-value items and n^H unique high-value items, respectively. Moreover, if $m^{-s}(n^L, n^H)$ and $m(n^L, n^H)$ denote the expected number of purchases by a given consumer type to draw n^L unique low-value items and n^H unique high-value items without and with the special item, respectively, so it must be the case that $m(n^L, n^H) = \frac{m^{-s}(n^L, n^H)}{1-q}$. Putting all the pieces together, we get

$$\lim_{n\to\infty} \frac{\Pi_B^*(n)}{n} \le \lim_{n\to\infty} \frac{\Pi_B^{-s}(n)}{n(1-q)} \le \frac{1}{e} \max\{\omega(v^H - c) + (1-\omega)(v^L - c),\ \omega(v^H - c)\}.$$

In the last step, we use the fact that $\lim_{n\to\infty} 1 - q \to 1$.

Now that we have an upper bound on the profit under blind box when $n \to \infty$, we consider separate selling. Specifically, we begin by defining a proxy solution whose profit is close to that of the optimum solution and compare that to Π_B^*. Formally, let Π_S^D denote the seller's profit under separate selling when the seller can set a personalized price for each consumer, and specifically prices the items as follows:

- If $c \le v^L$: For any given consumer, the price for regular item $i \in [n]$ equals the consumer's valuation for the item, i.e., $v_i \in \{v^L, v^H\}$. Furthermore, the special item is priced at zero, i.e., $p_s = x_s = 0$,

- If $c > v^L$: For any given consumer, the price for regular item $i \in [n]$ is given by $v_i = 0$ if it is a low-valued regular item and $v_i = v^H$ if it is a high-valued regular item. Furthermore, the special item is priced at zero, i.e., $p_s = x_s = 0$.

We reiterate that this hypothetical solution is only useful as a proxy for comparing profits. Recall that consumer types are characterized by (ℓ_r, ℓ_s) where ℓ_r denotes the number of regular items that the consumer values at $v_i = v^H$ and $l_s = 1$ implies that the consumer's base reward for the special item is given by $v_0 = v^H$. Let $\Pi_S^D(\ell_r, \ell_s)$ represent the profit under the proxy solution from consumers of type (ℓ_r, ℓ_s). Then, $\Pi_S^D(\ell_r, \ell_s) = (n - \ell_r)(v^L - c) + \ell_r(v^H - c)$ if $c \le v^L$ and $\Pi_S^D(\ell_r, \ell_s) = \ell_r(v^H - c)$ if $c \in (v^L, v^H)$.

Taking expectation over ℓ_r, ℓ_s for the case where $c \le v^L$ yields

$$\Pi_S^D(n) = \sum_{\ell_r=0}^{n} \sum_{\ell_s=0}^{1} \mathbb{P}(\ell_r, \ell_s) \left((n - \ell_r)(v^L - c) + \ell_r(v^H - c)\right) = n\left((1-\omega)v^L + \omega v^H - c\right), \tag{C.1}$$

where $\mathbb{P}(\ell_r, \ell_s) = \omega^{\ell_r + \ell_s}(1-\omega)^{n+1-\ell_r-\ell_s}$. Similarly, when $c > v^L$, we get that $\Pi_S^D(n) = n\omega(v^H - c)$.

In the following, we first show that $\frac{\Pi_B^*(n)}{\Pi_S^D(n)} \le 1/e$ for all $c \in [0, v^H]$. Following this, we compare the profit under the proxy solution, i.e., $\Pi_S^D(n)$, to the actual profit under separate selling, i.e., $\Pi_S^*(n)$, in order to complete the proposition.

First, from equations (E.4) and (C.1), when $c \le v^L$, we have

$$\lim_{n\to\infty} \frac{\Pi_B^*(n)}{\Pi_S^D(n)} \le \lim_{n\to\infty} \frac{n\frac{(1-\omega)(v^L-c)+\omega(v^H-c)}{e}}{n((1-\omega)(v^L-c)+\omega(v^H-c))} = \frac{1}{e}.$$

Similarly, when $c > v^L$, we have

$$\lim_{n\to\infty} \frac{\Pi_B^*(n)}{\Pi_S^D(n)} \le \lim_{n\to\infty} \frac{\frac{n\omega(v^H-c)}{e}}{n\omega(v^H-c)} = \frac{1}{e}.$$

Therefore, we can conclude that $\Pi_B^*/\Pi_S^D \le 1/e$ for all $c \in [0, v^H]$. To complete the proof, we now compare the proxy solution to the actual profit under separate selling. In particular, we show that $\Pi_S^D/\Pi_S^* \le 2$ for all n. One can obtain a simple lower bound for the optimal profit under separate selling by considering two solutions: (i) pricing all regular items at $p_r = v^L$, (ii) pricing all regular items at $p_r = v^H$. In both cases, we ignore profits from the special item (e.g., by not producing any quantity of said item). Taking a lower bound based on the two proposed solutions, we get

$$\Pi_S^* \ge n \max\left(v^L - c, \omega(v^H - c)\right).$$

We now compare the above lower bound with the seller's profit under the proxy solution. Namely, for any given $n \ge 1$, we have

$$\frac{\Pi_S^D}{\Pi_S^*} \le \frac{n\max\{(1-\omega)(v^L-c)+\omega(v^H-c), \omega(v^H-c)\}}{n\max\{(v^L-c), \omega(v^H-c)\}} = \frac{\max\{(1-\omega)(v^L-c)+\omega(v^H-c), \omega(v^H-c)\}}{\max\{(v^L-c), \omega(v^H-c)\}}.$$

Once again, we consider two cases. When $c < v^L$, we have

$$\frac{\Pi_S^D}{\Pi_S^*} \leq \frac{(1-\omega)(v^L-c)+\omega(v^H-c)}{\max\{(v^L-c),\omega(v^H-c)\}} \leq \frac{(1-\omega)(v^L-c)+\omega(v^H-c)}{\frac{1}{2}\left((1-\omega)(v^L-c)+\omega(v^H-c)\right)} \leq 2.$$

In the second inequality, we use the fact that $\max(a,b) \geq \frac{1}{2}(a+b)$ when all of the parameters are positive. On the other hand, when $c > v^L$, we get

$$\frac{\Pi_S^D}{\Pi_S^*} \leq \frac{\omega(v^H-c)}{\omega(v^H-c)} = 1.$$

Combining both arguments, we conclude $\Pi_B^*/\Pi_S^* \leq 2/e < 1$. □

C.2. Proofs from Section 4

Proof of Lemma 4.

Proof of Statement (i). Given $\alpha > \frac{4}{\omega^2}$, we first establish that $\Pi_B^* > \Pi_S^*$ if $n \leq \underline{n}$, where the threshold $\underline{n} \geq 1$. By Proposition 2, for any fixed n, we have $\Pi_B^*(n) \geq \Pi_S^*(n)$ if $\alpha > \frac{4n}{\omega^2}$. Therefore, it follows that $\Pi_B^*(n=1) > \Pi_S^*(n=1)$ if $\alpha > \frac{4}{\omega^2}$. Given α, let $\underline{n}$ denote the largest value of n such that $\Pi_B^*(n) \geq \Pi_S^*(n)$ for all $n \leq \underline{n}$. The first statement in the lemma follows. Note that $\underline{n}$ is not vacuous since we know that blind boxes dominate at $n=1$ when $\alpha > \frac{4}{\omega^2}$.

Proof of Statement (ii). Proposition 2 shows that $\Pi_B^* < \Pi_S^*$ as $n \to \infty$ for any given $\alpha \geq 0$. Therefore, there exists a threshold $\bar{n}$ such that for all $n \geq \bar{n}$, $\Pi_B^* \leq \Pi_S^*$ holds for all $\alpha \geq 0$. □

Proof of Proposition 3. For any given $n \geq 1$, the profit for blind boxes with parameters (p_b, q) is given by

$$\dot{\Pi}_B(p_b,q;n) = \Pi_B(p_b,q;n) - C(n) - K = m(p_b-c) - C(n) - K,$$

where m is the expected number of blind boxes sold, $C(n)$ is the fixed cost for n regular items and K is the fixed cost for the special item. The profit under separate selling with an endogenous number of items is given by

$$\begin{aligned}\dot{\Pi}_S(k,x_s;n) &= \Pi_S(k,x_s;n) - C(n) - K \\ &= k(v^L-c) + (n-k)\omega(v^H-c) + x_s\{(1+\alpha[1-r(k,x_s)])v_0 - c\} - C(n) - K.\end{aligned}$$

As introduced in Section 4.1, we assume that there exists a constant $n_{\max}$ such that the seller would never manufacture more than $n_{\max}$ items under either scheme, as the cost would become prohibitively high.

Proof of Statement (i). We show that if $\alpha \geq \dot{\alpha}_1$, then the optimal blind box profit $\dot{\Pi}_B^*$ is strictly decreasing in n, implying $n_B^* = 1$. As a corollary, it follows that $n_B^* \leq n_S^*$.

Since $C(n)$ is strictly increasing in n, the monotonicity of $\dot{\Pi}_B^*$ in n is determined by the monotonicity of Π_B^*, which denotes the optimal blind box profit without fixed costs. Hence, it suffices to prove that $\Pi_B^*(n)$ is strictly decreasing in n once α exceeds $\dot{\alpha}_1$. By Lemma E2, for any $\beta > 0$ and each n there exists a threshold $\tilde{\alpha}^H(n)$ such that if $\alpha \geq \tilde{\alpha}^H(n)$, the optimal pricing takes the form

$$p_b^* = q^* v_s(q^*, v_0), \qquad q^* \in \arg\max_{q \in (0, \frac{1}{n+1}]} \left\{ v_s(q, v_0) - \frac{c}{q} \right\}.$$

Let $\Pi_B^H(n)$ denote the profit when $v_0 = v^H$ and $\Pi_B^L(n)$ when $v_0 = v^L$. Then, we have

$$\Pi_B^H(n) = (1+\alpha)v^H - c - 2\sqrt{n\alpha v^H c} \quad \text{if } v_0 = v^H,$$

or

$$\Pi_B^L(n) = (1+\alpha)v^L - c - 2\sqrt{n\alpha v^L c} \quad \text{if } v_0 = v^L.$$

Let $\mathcal{N} = \{1, 2, \cdots, n_{\max}\}$ and define the threshold $\dot{\alpha}_1$ as $\dot{\alpha}_1 = \sup_{n \in \mathcal{N}} \tilde{\alpha}^H(n)$. Then for any $\alpha \geq \dot{\alpha}_1$ and all $n \in \mathcal{N}$, the optimal profit under blind box is $\Pi_B^*(n) = \max(\Pi_B^H(n), \Pi_B^L(n))$. Notably, both $\Pi_B^H(n)$ and $\Pi_B^L(n)$ are strictly decreasing in n. Therefore, it follows that the optimal profit $\Pi_B^*(n)$ is also strictly decreasing in n for all $\alpha \geq \dot{\alpha}_1$.

The proof of Statement (i) follows as $n_B^* = 1$.

Proof of Statement (ii). We next show that there exists a threshold $\dot{\alpha}_2$ such that $\dot{\Pi}_B^* > \dot{\Pi}_S^*$ if $\alpha > \dot{\alpha}_2$ under the optimal choice of n. By Proposition 2, for any given β and n, there exists a threshold $\alpha^H(n)$ such that if $\alpha > \alpha^H(n)$, then $\Pi_B^*(n) > \Pi_S^*(n)$. Recall that n_S^* denotes the optimal number of regular items chosen by the seller under separate selling. Applying Proposition 2 with $n = n_S^*$ yields that for any $\alpha > \alpha^H(n_S^*)$, we have

$$\Pi_B^*(n_S^*) > \Pi_S^*(n_S^*).$$

Subtracting the same fixed costs on both sides, we get

$$\dot{\Pi}_B(n_S^*) = \Pi_B^*(n_S^*) - C(n_S^*) - K > \Pi_S^*(n_S^*) - C(n_S^*) - K = \dot{\Pi}_S^*(n_S^*).$$

Finally, we note that the seller's profit under a blind box with n_S^* regular items is smaller than or equal to their profit under the same mechanism but with n_B^* items, as the latter quantity denotes the optimal choice n with the blind box. This implies $\dot{\Pi}_B^*(n_B^*) \geq \dot{\Pi}_B(n_S^*) > \dot{\Pi}_S^*(n_S^*)$ for all $\alpha > \alpha^H(n_S^*)$. In conclusion, there exists a threshold $\dot{\alpha}_2 \leq \alpha^H(n_S^*)$ such that $\dot{\Pi}_B^* > \dot{\Pi}_S^*$ if $\alpha > \dot{\alpha}_2$. This completes the proof. □

Proof of Corollary 2. This follows directly from the proof of Proposition 3, Statement (ii). □

Proof of Proposition 4. By Proposition 2, we know that in the base model (where α is homogeneous), the optimal profit under blind box, Π_B^*, exceeds the profit under separate selling, Π_S^*, if α is large. We now prove that this result continues to hold in the setting where α is heterogeneous. Recall that consumers' valuation for the special item is given by

$$v(s) = \begin{cases} v_0(1+\alpha_L(1-r)) & \text{with probability } \frac{1}{2}, \\ v_0(1+\alpha_H(1-r)) & \text{with probability } \frac{1}{2}, \end{cases}$$

where $v_0 = v^H$ with probability ω and $v_0 = v^L$ with probability $1-\omega$. Under the optimal profit, we then have $p_s = v_0(1+\alpha_L(1-r))$ or $v_0(1+\alpha_H(1-r))$, for some suitable value of r. Any value of p_s that does not align with this form is clearly suboptimal as the seller can simply increase the price to $p_s \in \{v_0(1+\alpha_L(1-r)), v_0(1+\alpha_H(1-r))\}$ for the same choice of r. Let $\hat{\Pi}(\alpha_L)$ and $\hat{\Pi}(\alpha_H)$ denote the profits under those two price forms, respectively. Then,

$$\hat{\Pi}_S^* = \max\{\hat{\Pi}(\alpha_L), \hat{\Pi}(\alpha_H)\}.$$

We will show that for any $\epsilon > 0$ and β, $\Pi_B^* > \hat{\Pi}(\alpha_L)$ and $\Pi_B^* > \hat{\Pi}(\alpha_H)$ for sufficiently large α, which implies $\Pi_B^* > \hat{\Pi}_S^*$.

Case (i): $p_s = v_0(1+\alpha_L(1-r))$. For separate selling, the heterogeneous case can be viewed as a special case of the base model with $\alpha = \alpha_L = \alpha_0 - \epsilon$ for any given $\epsilon \in [0, \alpha_0]$, and the seller needs to further optimize $p_r(i)$, v_0, and r. Then, we have $\hat{\Pi}_S^*(\alpha_L) = \Pi_S^*(\alpha = \alpha_0 - \epsilon)$. Then, by the same argument used at the beginning of this proof, there exist thresholds $\hat{\alpha}$ such that $\Pi_B^*(\alpha = \alpha_0 - \epsilon) > \hat{\Pi}_S^*(\alpha_L)$ if $\alpha_0 > \hat{\alpha}$. It remains to show that the optimal blind box profit under heterogeneity weakly dominates that of the base model:

$$\hat{\Pi}_B^* \geq \Pi_B^*(\alpha = \alpha_0 - \varepsilon).$$

Fix the blind box parameters (p_b^*, q^*) that are optimal for the base model with $\alpha = \alpha_0 - \varepsilon$. Let m_{hom} be the corresponding expected purchase quantity in the base model, and let m_{het} be the expected purchase quantity under the heterogeneous case. Because a higher status parameter weakly increases the expected value of buying another box (if the special item is not collected), no consumer with α_H stops earlier than the same type in the homogeneous α_L model. Therefore,

$$m_{\text{het}}(p_b^*, q^*) \geq m_{\text{hom}}(p_b^*, q^*).$$

Then, the profit is weakly increasing at the fixed (p_b^*, q^*). Hence,

$$\hat{\Pi}_B(p_b^*, q^*) = (p_b^* - c)m_{\text{het}}(p_b^*, q^*) \geq (p_b^* - c)m_{\text{hom}}(p_b^*, q^*) = \Pi_B(p_b^*, q^*).$$

Maximizing over (p_b, q) on the left yields $\hat{\Pi}_B^* \geq \hat{\Pi}_B(p_b^*, q^*)$, as claimed.

Case (ii): $p_s = v_0\big(1+\alpha_H(1-r)\big)$**.** We derive a lower bound for the blind box profit and show that it exceeds the upper bound for the separate selling profit if α_H is large.

Consider a blind box in which one-half of consumers have a high status-seeking level α_H, while the remaining half do not purchase. By Lemma E2, if $\alpha_H = \alpha_0 + \epsilon > \tilde{\alpha}^H$, then

$$\hat{\Pi}_B \geq \frac{1}{2}((1+\alpha_H)v^L - c - 2\sqrt{n\alpha v^L c}) \quad \text{if } p_b = qv_s(v_0 = v^L),$$

and

$$\hat{\Pi}_B \geq \frac{\omega}{2}((1+\alpha_H)v^H - c - 2\sqrt{n\alpha v^H c}) \quad \text{if } p_b = qv_s(v_0 = v^H).$$

The factor 1/2 arises because only half of the consumers are highly status-seeking. Hence,

$$\hat{\Pi}_B^* \geq \max\left\{\frac{1}{2}((1+\alpha_H)v^L - c - 2\sqrt{n\alpha v^L c}), \frac{\omega}{2}((1+\alpha_H)v^H - c - 2\sqrt{n\alpha v^H c})\right\}.$$

Under separate selling,

$$\hat{\Pi}_S^* = k^*(v^L - c) + (n-k^*)(\omega v^H - c) + x_s^*\left(v_0^*\big(1+\alpha_H(1-r(k^*,x_s^*))\big) - c\right), \quad r(k^*, x_s^*) = \frac{x_s^* n}{k^* + (n-k^*)\omega}.$$

We analyze the cases $v_0^* = v^H$ and $v_0^* = v^L$, and show that $\hat{\Pi}_B^* \geq \hat{\Pi}_S^*$ once α_H exceeds a threshold $\hat{\alpha}_H$.

If $v_0^* = v^H$, then we have

$$\begin{aligned}
\hat{\Pi}_S^* &= k^*(v^L - c) + (n-k^*)(\omega v^H - c) + x_s^*\big(\big(1+\alpha_H(1-x_s^*)\big)v^H - c\big) \\
&\leq n(v^H - c) + \frac{\omega}{2}(v^H - c) + \frac{\omega}{2}(1-\frac{\omega}{2})\alpha_H v^H \\
&\leq (n+1)(v^H - c) + \frac{\omega}{2}(1-\frac{\omega}{2})\alpha_H v^H,
\end{aligned}$$

where the first inequality uses the bound $x_s^* \leq \frac{\omega}{2} \leq \frac{1}{2}$ and the fact that $x_s(1-x_s)$ is concave and maximized over $[0, \omega/2]$ at $x = \omega/2$.

For $\hat{\Pi}_B^* > \hat{\Pi}_S^*$, it suffices that

$$(1+\alpha_H)v^H - c - 2\sqrt{n\alpha_H v^H c} > (n+1)(v^H - c) + \Big(\frac{\omega}{2} - \frac{\omega^2}{4}\Big)\alpha_H v^H.$$

which is equivalent to

$$v^H\Big(1 - \frac{\omega}{2} + \frac{\omega^2}{4}\Big)\alpha_H - 2\sqrt{n\alpha_H v^H c} > n(v^H - c). \tag{C.2}$$

The right-hand side of (C.2) is independent of α_H, while the left-hand side increases in α_H if $\alpha_H > 16nc/(9v^H)$, i.e., the derivative is positive. Hence there exists a threshold $\hat{\alpha}_H$ such that $\hat{\Pi}_B^* > \hat{\Pi}_S^*$ for all $\alpha_H > \hat{\alpha}_H$.

If $v_0^* = v^L$, then we have

$$\begin{aligned}\hat{\Pi}_S^* &= k^*(v^L - c) + (n - k^*)(\omega v^H - c) + x_s^*\big((1 + \alpha_H(1 - x_s^*)\big)v^L - c) \\ &\le n(v^H - c) + (v^H - c) + x_s^*(1 - x_s^*)\alpha_H v^L \\ &\le (n+1)(v^H - c) + \frac{1}{4}\alpha_H v^L,\end{aligned}$$

where the second inequality follows by maximizing the concave quadratic $x_s(1-x_s)$ over $x_s \in [0,1]$.

For $\hat{\Pi}_B^* > \hat{\Pi}_S^*$, it suffices that

$$\frac{1}{2}((1+\alpha_H)v^L - c - 2\sqrt{n\alpha_H v^L c}) > (n+1)(v^H - c) + \frac{1}{4}\alpha_H v^L,$$

which is equivalent to showing

$$\frac{1}{2}\alpha_H v^L + \frac{1}{2}v^L - c - 2\sqrt{n\alpha_H v^L c} > (n+1)(v^H - c).$$

The left-hand side is eventually increasing in α_H if $\alpha_H > 16nc/v^L$, i.e., the derivative is positive, so there exists a threshold $\hat{\alpha}_H$ such that $\hat{\Pi}_B^* > \hat{\Pi}_S^*$ for all $\alpha_H > \hat{\alpha}_H$.

Finally, let $\hat{\alpha} = \max\{\tilde{\alpha}^H, \hat{\alpha}_H\} - \epsilon$. Then $\hat{\Pi}_B^* > \hat{\Pi}_S^*$ holds for all $\alpha_0 > \hat{\alpha}$, which completes the proof. □

Proof of Corollary 3. We tackle the proof in three steps. First, we establish the existence of two thresholds $\hat{\alpha}_b$ and $\hat{\epsilon}_1$ such that if $\alpha_0 > \hat{\alpha}_b$ and $\epsilon > \hat{\epsilon}_1$, the optimal price under blind box is given by $p_b = qv_s(v_0, \alpha_H)$ for $v_0 \in \{v^L, v^H\}$. Second, we show the optimal policy under separate selling if $\alpha_0 > \hat{\alpha}_s$ and $\epsilon > \hat{\epsilon}_1$. Finally, we demonstrate that as long as $\omega v^H > v^L$ and $\alpha_0 > \hat{\alpha}_1$, we have $\hat{\Pi}_B^* - \hat{\Pi}_S^*$ is strictly increasing in $\epsilon \in [\hat{\epsilon}_1, \alpha_0]$.

Step 1. We first show that any price $p_b \le (1+\alpha_L)v^H$ cannot be optimal for the blind box if α_0 is sufficiently large and $\epsilon = \alpha_0 - \epsilon_\alpha$. At $\epsilon = \alpha_0 - \epsilon_\alpha$, $\alpha_L = \epsilon_\alpha$ and $\alpha_H = 2\alpha_0 - \epsilon_\alpha$. Equivalently, it suffices to show that no price $p_b \le (1+\alpha_L)v^H$ can be optimal if α_H is sufficiently large.

Note that a consumer with a low status-seeking level α_L will not make any purchase if $p_b > (1+\alpha_L)v^H$. Specifically, they do not purchase any blind box as their value from the first purchase is at most $(1-q)v^H + q(1 + (1 - \frac{nq}{1-q})\alpha_L)v^H$, which is strictly below $(1+\alpha_L)v^H$.

Next, we construct a lower bound on the optimal profit under the blind box via the following price:

$$p_b = qv_s(v^L, \alpha_H) = \frac{v^L\left(1 + \alpha_H - \sqrt{\frac{nc\alpha_H}{v^L}}\right)}{1 + \sqrt{\frac{n\alpha_H v^L}{c}}}, \quad \text{with} \quad q = \frac{1}{1 + \sqrt{n\alpha_H v^L/c}}.$$

One can verify that $qv_s(v^L, \alpha_H)$ increases in α_H if $\alpha_H > (v^L + c)/c$, and α_L decreases in α_H because $\alpha_L = 2\alpha_0 - \alpha_H$. Hence, there exists a threshold α_1 such that $qv_s(v^L, \alpha_H) > (1 + \alpha_L)v^H$ for all $\alpha_H > \alpha_1$. Consequently,

$$\hat{\Pi}_B(p_b, q) \geq \frac{1}{2}\Big((1+\alpha_H)v^L - c - 2\sqrt{n\alpha_H v^L c}\Big), \tag{C.3}$$

where the inequality holds due to the fact that consumers with $\alpha = \alpha_H$ may keep purchasing even after obtaining the special item, while consumers with $\alpha = \alpha_L$ make no purchases.

We now show that for sufficiently large α_H, any price $p_b' \leq (1+\alpha_L)v^H$ yields strictly lower profit than (C.3). Consider any blind box (p_b', q') with $p_b' \leq (1+\alpha_L)v^H$. Then, we have

- For consumers with $\alpha = \alpha_L$, the expected number of purchases is at most $\frac{n(1+\log n)}{1-q'}$ if $p_b' > q'v_s(v^H, \alpha_L)$, and at most $\frac{n(1+\log n)}{1-q'} + \frac{1}{q'}$ if $p_b' \leq q'v_s(v^H, \alpha_L)$.
- For consumers with $\alpha = \alpha_H$, the expected number of purchases is at most $\frac{n(1+\log n)}{1-q'}$ if $p_b' > q'v_s(v^L, \alpha_H)$, and at most $\frac{n(1+\log n)}{1-q'} + \frac{1}{q'}$ if $p_b' < q'v_s(v^L, \alpha_H)$.

Moreover, for any $0 < v^L \leq v^H$ and fixed $0 < q' < 1/(n+1)$, we obtain

$$2q'v_s(v^H, \alpha_L) = 2q'\Big(1 + \Big(1 - \frac{nq'}{1-q'}\Big)\alpha_L\Big)v^H < q'v_s(v^L, \alpha_H) = q'\Big(1 + \Big(1 - \frac{nq'}{1-q'}\Big)\alpha_H\Big)v^L, \tag{C.4}$$

if $\alpha_H > \alpha_2 = \sup_{q'}\{\alpha_2(q')\}$.[9]

We demonstrate that there exists a threshold α_3 such that $\hat{\Pi}_B(p_b', q') < \hat{\Pi}_B(p_b, q)$ as long as $\alpha > \alpha_3$. We prove this in three cases: (1) $p_b' > q'v_s(v^L, \alpha_H)$, (2) $q'v_s(v^H, \alpha_L) < p_b' \leq q'v_s(v^L, \alpha_H)$ and (3) $p_b' \leq q'v_s(v^H, \alpha_L)$.

(1) $p_b' > q'v_s(v^L, \alpha_H)$. In this case, both consumer types purchase at most $\frac{n(1+\log n)}{1-q'}$ boxes, so

$$\hat{\Pi}_B(p_b', q') \leq \frac{n(1+\log n)}{1-q'}((1+\alpha_L)v^H - c).$$

The right-hand side is strictly decreasing in α_H, whereas the lower bound in (C.3) is strictly increasing in α_H. Therefore, there exists α_3 such that any price $p_b' \geq q'v_s(v^L, \alpha_H)$ is strictly dominated.

(2) $q'v_s(v^H, \alpha_L) < p_b' \leq q'v_s(v^L, \alpha_H)$. In this region, the profit can be written as:

$$\begin{aligned}\hat{\Pi}_B(p_b', q') &\leq \frac{n\big(1+\log n\big)}{2(1-q')}((1+\alpha_L)v^H - c) + \left(\frac{n(1+\log n)}{2(1-q')} + \frac{1}{2q'}\right)\big(\min\{(1+\alpha_L)v^H, q'v_s(v^L, \alpha_H)\} - c\big) \\ &= \frac{n(1+\log n)}{(1-q')}((1+\alpha_L)v^H - c) + \frac{1}{2q'} \cdot \big(\min\{(1+\alpha_L)v^H, q'v_s(v^L, \alpha_H)\} - c\big).\end{aligned}$$

[9] If $q' \to 1/(n+1)$, the above inequality may fail to hold for any finite $\alpha_2(q')$. However, in this limit we have $v_s(v^L, \alpha_H) \to v^L$ and $v_s(v^H, \alpha_L) \to v^H$. In that case, there is no special item and the profit is bounded and $\hat{\Pi}_B' \leq (n+1)(1+\log n)(v^H - c)$, which is dominated by $\hat{\Pi}_B$ because $\hat{\Pi}_B$ grows in α_H. Thus, the dominance argument remains valid.

Let $\tilde{q}$ solve $q'v_s(v^L,\alpha_H)=(1+\alpha_L)v^H$. We show that, for sufficiently large α_H, the following inequality holds:

$$\frac{1}{2q'}\Big(\min\{(1+\alpha_L)v^H, q'v_s(v^L,\alpha_H)\}-c\Big)\le\frac{1}{2\tilde{q}}((1+\alpha_L)v^H-c).$$

Define

$$z_1(q')=q'v_s(v^L,\alpha_H)-(1+\alpha_L)v^H=q'\Big(1+\Big(1-\frac{nq'}{1-q'}\Big)\alpha_H\Big)v^L-(1+\alpha_L)v^H.$$

The function $z_1(q')$ increases on $\Big(0,\frac{\sqrt{\alpha_H n}}{\sqrt{\alpha_H n+\alpha_H+1}}\Big)$ and decreases on $\Big(\frac{\sqrt{\alpha_H n}}{\sqrt{\alpha_H n+\alpha_H+1}},\frac{1}{n+1}\Big)$, and $\frac{\sqrt{\alpha_H n}}{\sqrt{\alpha_H n+\alpha_H+1}}<\frac{1}{n+1}$ if $\alpha_H>\frac{n}{n+1}$. Thus, $z_1(q')$ is unimodal on $\Big(0,\frac{1}{n+1}\Big)$. Moreover,

$$z_1(0)=-(1+\alpha_L)v^H<0,\qquad z_1\left(\frac{1}{n+1}\right)=\frac{v^L}{n+1}-(1+\alpha_L)v^H<0,$$

and

$$z_1\left(\frac{1}{2n+1}\right)=\frac{1}{2n+1}\Big(1+\frac{\alpha_H}{2}\Big)v^L-(1+\alpha_L)v^H.$$

For sufficiently large α_H, $z_1(\frac{1}{2n+1})>0$, so there exist $q_1'<q_2'$ such that $z_1(q_1')=z_1(q_2')=0$. Hence,

$$q'v_s(v^L,\alpha_H)>(1+\alpha_L)v^H\quad\text{for all } q'\in(q_1',q_2'),$$

and q_1',q_2' satisfy $q'v_s(v^L,\alpha_H)=v^H$. Then, we have $\tilde{q}=\tilde{q}_1$ since $\frac{1}{2\tilde{q}}(v^H-c)$ decreases with $\tilde{q}$.

Next define

$$z_2(q')=v_s(v^L,\alpha_H)-\frac{c}{q'},$$

which is unimodal and achieves its maximum at $q^*=\frac{1}{1+\sqrt{n\alpha_H v^L/c}}$. For large α_H, $q_1'<q^*<q_2'$ because

$$z_1(q^*)=v^L\frac{1+\alpha_H-\sqrt{\alpha_H n}\sqrt{\frac{c}{v^L}}}{1+\sqrt{\alpha_H n}\sqrt{\frac{v^L}{c}}}-v^H>0,$$

and $z_1(q_1')=z_1(q_2')=0$. Thus, $z_2(q')$ increases on $(0,q_1')$ and decreases on $\Big(q_2',\frac{1}{n+1}\Big)$

We now consider three cases based on the value of q'.

Case 1: $q'\le q_1'$. Then $\min(v^H,q'v_s)=q'v_s$, so

$$\begin{aligned}&\frac{1}{2q'}(\min\{(1+\alpha_L)v^H,q'v_s\}-c)\\=&\frac{1}{2}\Big(v_s(v^L,\alpha_H)-\frac{c}{q'}\Big)\\\le&\frac{1}{2}\Big(v_s(v^L,\alpha_H)-\frac{c}{q_1'}\Big)\\=&\frac{1}{2\tilde{q}}((1+\alpha_L)v^H-c),\end{aligned}$$

where the inequality uses that $z_2(q')$ is increasing on $(0, q_1')$, and the last equality follows from $\tilde{q} = \tilde{q}_1 = q_1'$.

Case 2: $q' \geq q_2'$. Similarly,

$$\begin{aligned}
&\frac{1}{2q'} \cdot \big(\min\{(1+\alpha_L)v^H, q' v_s(v^L, \alpha_H)\} - c\big) \\
=&\frac{1}{2} \cdot (v_s(v^L, \alpha_H) - \frac{c}{q'}) \\
\leq&\frac{1}{2} \cdot (v_s(v^L, \alpha_H) - \frac{c}{q_2'}) \\
\leq&\frac{1}{2\tilde{q}} \cdot ((1+\alpha_L)v^H - c),
\end{aligned}$$

where the first inequality holds since $z_2(q')$ is decreasing in $q' \in (q_2', \frac{1}{n+1})$. The second inequality arises from the fact that $\tilde{q} \geq q_2'$

Case 3: $q_1' < q' < q_2'$. Then $\min(v^H, q' v_s) = v^H$, so

$$\begin{aligned}
&\frac{1}{2q'} \cdot \big(\min\{(1+\alpha_L)v^H, q' v_s(v^L, \alpha_H)\} - c\big) \\
=&\frac{1}{2q'}((1+\alpha_L)v^H - c) \\
\leq&\frac{1}{2q_1'}((1+\alpha_L)v^H - c) \\
=&\frac{1}{2\tilde{q}_1}((1+\alpha_L)v^H - c),
\end{aligned}$$

where the inequality uses $q_1' < q'$, and the final equality due to the fact that $\tilde{q} = \tilde{q}_1 = q_1'$.

Combining all three cases yields the desired inequality.

Moreover, we have the expression of $\tilde{q}$ as

$$\tilde{q} = \frac{\big(v^L(1+\alpha_H) + (1+\alpha_L)v^H\big) - \sqrt{\big(v^L(1+\alpha_H) + (1+\alpha_L)v^H\big)^2 - 4v^L v^H\big(1+\alpha_H(n+1)\big)\big(1+\alpha_L\big)}}{2v^L\big(1+\alpha_H(n+1)\big)}.$$

Then, as $\alpha_H \to \infty$, we have

$$\lim_{\alpha_H \to \infty} \frac{\hat{\Pi}_B(p_b', q')}{\alpha_H} \leq \lim_{\alpha_H \to \infty} \frac{1}{2\tilde{q}\alpha_H}((1+\alpha_L)v^H - c) = \frac{1}{2}(v^L - \frac{cv^L}{(1+\alpha_L)v^H}).$$

For any $c > 0$, the above expression is strictly smaller than $\lim_{\alpha_H \to \infty} \frac{\Pi_B\left(p_b, q\right)}{\alpha_H} \geq \frac{v^L}{2}$. [10]

Therefore, there exists a cutoff α_3 such that any blind box price $q' v_s(v^H, \alpha_L) < p_b' \leq q' v_s(v^L, \alpha_H)$ is strictly suboptimal.

[10] Note that $c = 0$, one can prove this separately as in the proof of Lemma E2. We do not repeat the steps here for brevity.

1. Suppose that $p_b' \le q' v_s(v^H, \alpha_L)$. Then, we have

$$\begin{aligned}
\hat{\Pi}_B(p_b', q') &\le (\frac{n(1+\log n)}{(1-q')} + \frac{1}{q'})(q' v_s(v^H, \alpha_L) - c) \\
&\le \cdot \left(\frac{n(1+\log n)}{2(1-q')} + \frac{1}{2q'}\right)\left(2q' v_s(v^H, \alpha_L) - c\right) \\
&\le \left(\frac{n(1+\log n)}{2(1-q')} + \frac{1}{2q'}\right)\left(q' v_s(v^L, \alpha_H) - c\right) \\
&\le 2 \cdot \left(\frac{n(1+\log n)}{2(1-q')} + \frac{1}{2q'}\right)\left(\min\{(1+\alpha_L)v^H, q' v_s(v^L, \alpha_H)\} - c\right) \\
&\le \hat{\Pi}_B(p_b, q),
\end{aligned}$$

where the first inequality holds because $2q' v_s(v^H, \alpha_L) \le q' v_s(v^L, \alpha_H)$ by (C.4), and the second and third inequalities follow by a similar argument as in Case 2

Therefore, there exists a cutoff α_3 such that any blind box price $p_b' \le q' v_s(v^H, \alpha_L)$ is strictly suboptimal.

In summary, we shown that at $\epsilon = \alpha_0 - \epsilon_\alpha$, any price $p_b \le (1+\alpha_L)v^H$ is suboptimal if $\alpha_H > \max\{\alpha_1, \alpha_2, \alpha_3\}$. Then, we can infer this result holds for a cutoff $\hat{\epsilon}_1$ such that for all $\epsilon \in [\alpha_0 - \epsilon_\alpha, \alpha_0]$, any price $p_b \le (1+\alpha_L)v^H$ is suboptimal. That is, if consumers do not purchase at $p_b \le (1+\alpha_L)v^H = (1+\epsilon_0)v^H$, they will not purchase at any higher price. Meanwhile, the lower bound profit for $p_b > (1+\alpha_L)v^H$ is strictly increasing in ϵ.

Moreover, since consumers with α_L do not purchase even a single blind box if $p_b > (1+\alpha_L)v^H$, in this region the problem reduces to the base model in which a mass 1/2 of consumers all have the same status parameter $\alpha = \alpha_H$ for the special item. By Lemma E2, there exists a threshold $\tilde{\alpha}^H$ such that $p_b = q v_s(v_0, \alpha_H)$ is the optimal blind box price is $\alpha_H > \tilde{\alpha}^H$ for $v_0 \in \{v^H, v^L\}$. Combining the above, we conclude that there exist suitable cutoffs $\hat{\alpha}_b$ and $\hat{\epsilon}_1$ such that, for all $\alpha_0 > \hat{\alpha}_b$ and $\epsilon > \hat{\epsilon}_1$, the optimal blind box price is $p_b = q v_s(v_0, \alpha_H)$.

Step 2. We fixed $\epsilon = \alpha_0 - \epsilon_\alpha$ first and show that $p_s^* = (1 + (1-x_s^*)\alpha_H)v_0$. In the final step, we show that this result still holds for all $\epsilon \in [\alpha_0 - \epsilon_\alpha, \alpha_0]$. Recall that $v_s(v^H, \alpha_L) < v_s(v^L, \alpha_H)$ if $\alpha_H > \alpha_2$.[11]

The price of the special item under separate selling can therefore be written as

$$p_s = \begin{cases} (1+(1-x_s)\alpha_L)v^L & x_s \in [0,1], \\ (1+(1-x_s)\alpha_H)v^L & x_s \in [0,1/2], \\ (1+(1-x_s)\alpha_L)v^H & x_s \in [0,(1+\omega)/2], \\ (1+(1-x_s)\alpha_H)v^H & x_s \in [0,\omega/2]. \end{cases}$$

[11] By Step 1, $v_s(v^H, \alpha_L) < v_s(v^L, \alpha_H)$ holds for any q', where $r = nq'/(1-q') \in (0,1)$. Under separate selling, the same inequality also holds for $r = x_s/(k+\omega(n-k)) \in (0,1)$. If $r = 0$, no special item is supplied; if $r = 1$, no special item exists.

We show that if $\alpha_H > \alpha_4$, the seller strictly prefers the α_H-based price over the α_L-based price. In particular, the profit under $(1+(1-x_s)\alpha_L)v^L$ is dominated by that under $(1+(1-x_s)\alpha_H)v^L$. Let (k', x_s') denote the optimal solution under $(1+(1-x_s)\alpha_L)v^L$. Then,

$$\begin{aligned}&k'(v^L-c)+(n-k')(v^H-c)+x_s'((1+(1-x_s')\alpha_L)v^L-c)\\ \le&k'(v^L-c)+(n-k')(v^H-c)+x_s((1+(1-x_s)\alpha_H)v_0-c),\end{aligned}$$

for some suitable x_s. Specifically, if $x_s' \le 1/2$, let $x_s = x_s'$, and the inequality holds since $\alpha_H > \alpha_L$. If $x_s' > 1/2$, let $x_s = 1/2$; the inequality still holds if $\alpha_H > \alpha_4$ for some suitable α_4 because the right-hand side is increasing in α_H, while the left-hand side is weakly decreasing in α_H. Moreover, this inequality also implies that the optimal supply of the special item must satisfy $x_s \le 1/2$, and the seller has no incentive to lower the price to induce purchases from type-α_L consumers. The similar argument shows that the price $(1+(1-x_s)\alpha_L)v^H$ is dominated by $(1+(1-x_s)\alpha_H)v^H$ for some suitable α_4.

In short, at $\epsilon = \alpha_0 - \epsilon_\alpha$, $p_s^* = (1+(1-x_s^*)\alpha_H)v_0$ if $\alpha_H > \max\{\alpha_2, \alpha_4\}$. Moreover, for any fixed α_0, the profit under the α_H-based price increases with ϵ, while the profit under the α_L-based price decreases with ϵ. Then, we have $p_s^* = (1+(1-x_s^*)\alpha_H)v_0$ if $\alpha_H > \max\{\alpha_2, \alpha_4\}$ and $\epsilon > \alpha_0 - \epsilon_\alpha$. Equivalently, $p_s^* = (1+(1-x_s^*)\alpha_H)v_0$ if $\alpha_0 = (\alpha_H + \epsilon_\alpha)/2 > \hat{\alpha}_s$ for some suitable $\hat{\alpha}_s$, and $\epsilon > \alpha_0 - \epsilon_\alpha$.

Step 3. In the final step, we show that as long as $\omega v^H > v^L$ and $\alpha_0 > \hat{\alpha}_1$, then $\hat{\Pi}_B^* - \hat{\Pi}_S^*$ is strictly increasing in $\epsilon \in [\hat{\epsilon}_1, \alpha_0]$.

Under separate selling, it is optimal for the seller to price the special item based on α_H:

$$p_s = \begin{cases}(1+(1-x_s)\alpha_H)v^L & x_s \in [0, 1/2],\\ (1+(1-x_s)\alpha_H)v^H & x_s \in [0, \omega/2].\end{cases}$$

By Lemma E1, when $\alpha_H > \alpha_n$, the optimal solution satisfies $k^* = n$,[12] and thus

$$\hat{\Pi}_S^* = n(v^L - c) + x_s^*((1+(1-x_s^*)\alpha_H)v_0 - c),$$

where $0 < x_s^* \le 1/2$ if $v_0^* = v^L$ and $0 < x_s^* \le \omega/2 \le 1/2$ if $v_0^* = v^H$.

Let x_s' be the optimal supply under $v_0 = v^L$. Consider the feasible choice $x_s = \omega x_s'$. Then

$$\begin{aligned}&n(v^L-c)+x_s\big((1+(1-x_s)\alpha_H)v^H-c\big)-\big[n(v^L-c)+x_s'\big((1+(1-x_s')\alpha_H)v^L-c\big)\big]\\ &=x_sv^H-x_s'v^L+\alpha_H\big(x_s(1-x_s)v^H-x_s'(1-x_s')v^L\big)\\ &\ge x_sv^H-x_s'v^L\\ &=x_s'(\omega v^H-v^L)>0,\end{aligned}$$

[12] Although the feasible range of x_s differs from the base model, this does not affect the validity of Lemma E1.

where the inequality uses $x_s \le x'_s < 1/2$ so that $x_s(1-x_s) \le x'_s(1-x'_s)$ and $\alpha_H \ge 0$, and the strict positivity follows from $\omega v^H > v^L$. Hence, any optimum under $v_0 = v^L$ is strictly dominated by a feasible policy under $v_0 = v^H$, so $v_0^* = v^H$.

Under the blind box,

$$\hat{\Pi}_B(v_0) = \begin{cases} \frac{\omega}{2}((1+\alpha_H)v^H - c - 2\sqrt{n\alpha_H v^H c}), & v_0 = v^H, \\ \frac{1}{2}((1+\alpha_H)v^L - c - 2\sqrt{n\alpha_H v^L c}), & v_0 = v^L. \end{cases}$$

Comparing $\hat{\Pi}_B(v^H)$ and $\hat{\Pi}_B(v^L)$ yields that $\hat{\Pi}_B(v^H) > \hat{\Pi}_B(v^L)$ if

$$\omega\big((1+\alpha_H)v^H - c - 2\sqrt{n\alpha_H v^H c}\big) - \big((1+\alpha_H)v^L - c - 2\sqrt{n\alpha_H v^L c}\big) > 0.$$

The left side is strictly increasing in α_H if

$$\alpha_H > \frac{4nc\big(\max\{0, \omega\sqrt{v^H} - \sqrt{v^L}\}\big)^2}{\big(\omega v^H - v^L\big)^2}.$$

Therefore there exists α_5 such that, if $\alpha_H > \alpha_5$, the blind box optimally takes $p_b^* = qv_s(v^H)$.

We now analyze the monotonicity of $\hat{\Pi}_B^* - \hat{\Pi}_S^*$ with respect to ϵ.

The profit difference is

$$\begin{aligned} \hat{\Pi}_B^* - \hat{\Pi}_S^* &= \frac{\omega}{2}((1+\alpha_H)v^H - c - 2\sqrt{n\alpha_H v^H c}) \\ &\quad - n(v^L - c) - x_s^*((1+(1-x_s^*)\alpha_H)v^H - c) \\ &= \left(\frac{\omega - 2x_s^*(1-x_s^*)}{2}\right)\alpha_H v^H - \omega\sqrt{n\alpha_H v^H c} + \left(\frac{\omega - 2x_s^*}{2}\right)(v^H - c) - n(v^L - c), \end{aligned}$$

where the coefficient of α_H, $\frac{\omega - 2x^*(1-x^*)}{2}$, is strictly positive since $x_s^* \le \omega/2$. Differentiating with respect to α_H,

$$\frac{\partial}{\partial \alpha_H}\big(\hat{\Pi}_B^* - \hat{\Pi}_S^*\big) = \Big(\tfrac{\omega}{2} - x_s^*(1-x_s^*)\Big)v^H - \frac{\omega}{2}\sqrt{\frac{nv^H c}{\alpha_H}}.$$

Thus the derivative is strictly positive if

$$\alpha_H > \alpha_5 = \frac{\omega^2 nc}{(\omega - 2x^*(1-x^*))^2 v^H}.$$

Then for any $\alpha_H > \hat{\alpha}_1 = \max\{\hat{\alpha}_b, \hat{\alpha}_s, \alpha_n, \alpha_5, \alpha_6\}$, then simultaneously $k^* = n$, the blind box optimally takes $v_0 = v^H$, and the derivative of $\hat{\Pi}_B^* - \hat{\Pi}_S^*$ with respect to α_H is strictly positive. Moreover, $v_s^* = v^H$ if $\omega v^H > v^L$.

Finally, since $\alpha_H = \alpha_0 + \epsilon$ and $\epsilon \in [\hat{\epsilon}_1, \alpha_0]$, if $\alpha_0 > \hat{\alpha}_1$ then $\alpha_H \ge \alpha_0 > \hat{\alpha}_1$ for all such ϵ. Therefore $\hat{\Pi}_B^* - \hat{\Pi}_S^*$ is strictly increasing in ϵ over $[\hat{\epsilon}_1, \alpha_0]$, as claimed. □

Proof of Proposition 5. We first outline the key steps involved in the proof. Our first step is to show that when the lump-sum bonus is sufficiently large, the seller maximizes profits under separate

selling by setting a low enough price that all consumer types end up obtaining all $n+1$ items. This observation allows us to derive a closed-form expression for profit under separate selling. We then note that this profit does not depend on α, whereas the profit under the blind box is weakly increasing in α, which implies the proposition. Recall that consumer types are indexed by (ℓ_r, ℓ_s), where ℓ_r denotes the number of regular items for which $v_i = v^H$, and $\ell_s = 1$ ($\ell_s = 0$) indicates that $v_0 = v^H$ ($v_0 = v^L$). For $\alpha \in [0, \alpha^{\max}]$, define the lower bound on the bonus:

$$v^+ \geq \check{v} = v^H \left((n+1)H_{n+1} + \frac{(n+1+\alpha^{\max})(n+1)}{(1-\omega)^{n+1}} \right),$$

where $\check{v} > \frac{v^H(n+1+\alpha)-(n+1)v^L}{(1-\omega)^{n+1}}$.

Proof of Statement (i).

Separate Selling. Fix any $\alpha \in [0, \alpha^{\max}]$. We first show by contradiction that any set of prices under which some consumer type does not obtain the bonus is strictly suboptimal when $v^+ > \check{v}$. Towards this end, consider prices $\check{p}_r(i)$ for regular item $i \in [n]$ and $\check{p}_s$ for the special item. Suppose that at least one type (ℓ_r, ℓ_s) does not obtain the bonus. Then, consumers of type $(0,0)$, who have the lowest valuations for all items, must also fail to obtain the bonus. Under such prices, the seller's profit satisfies

$$\begin{aligned}\check{\Pi}_S(\check{p}_r(i), \check{p}_s) &\leq \sum_{\substack{(\ell_r,\ell_s)\\ \neq(0,0)}} \mathbb{P}(\ell_r, \ell_s) \left[n(v^H - c) + v^H(1+\alpha) - c + v^+ \right] + \mathbb{P}(0,0) \left(n(v^H - c) + v^H(1+\alpha) - c \right) \\ &= n(v^H - c) + v^H(1+\alpha) - c + v^+(1-(1-\omega)^{n+1}). \end{aligned} \tag{C.5}$$

Here, $\mathbb{P}(\ell_r, \ell_s)$ is the probability of type (ℓ_r, ℓ_s), and $\mathbb{P}(0,0) = (1-\omega)^{n+1}$. The upper bound follows from the fact that under separate selling, no consumer's payment can exceed their maximum attainable valuation from the $n+1$ items, including the bonus. A consumer's maximal value for each regular item is v^H, and that for the special item is $v^H(1+\alpha)$.

We now compare the above profit with a policy in which all consumer types receive the bonus. It is straightforward to argue that, under separate selling, any set of prices in which all consumers receive the bonus is profit-equivalent. Therefore, consider $\check{p}_r^*(i) = \check{p}_s^* = v^L + \frac{v^+}{n+1}$ and $x_s = 1$. Under these prices, every consumer purchases all $n+1$ items, yielding the profit

$$\check{\Pi}_S(\check{p}_r^*(i), \check{p}_s^*) = (n+1)(v^L - c) + v^+. \tag{C.6}$$

Comparing the above expression to Equation (C.5), we get that $\check{\Pi}_S(\check{p}_r^*(i), \check{p}_s^*) > \check{\Pi}_S(\check{p}_r(i), \check{p}_s)$ holds as long as

$$n(v^H - c) + v^H(1+\alpha) - c - (n+1)(v^L - c) < v^+(1-\omega)^{n+1}.$$

Simplifying, we get

$$v^+ > \frac{v^H(n+1+\alpha)-(n+1)v^L}{(1-\omega)^{n+1}}. \tag{C.7}$$

We note that the expression on the right-hand side above is at most $\check{v}$ since $\alpha \leq \alpha^{\max}$. In other words, as long as $v^+ > \check{v}$, we have the optimal profit under separate selling is $(n+1)(v^L - c) + v^+$. Notably, this expression holds for all $\alpha \in [0, \alpha^{\max}]$ and therefore, the profit under separate selling is independent of α in this range.

Blind Box. We argue that the seller's profit under the blind box is weakly increasing in α. Let $(\check{p}_b, \check{q})$ be the pair of blind box parameters that maximize profit at some $\alpha = \alpha_1$. Consider any $\alpha_2 > \alpha_1$ in the range $[0, \alpha^{\max}]$. Then, the expected number of blind boxes purchased by consumers under α_2 given $(\check{p}_b, \check{q})$ is at least the number of blind boxes purchased under $\alpha = \alpha_1$. To see why, note that for any consumer type (ℓ_r, ℓ_s) and any given state (n_r^H, n_r^L, n_s) denoting the number of high-value regular, low-value regular, and special items collected by the consumer respectively, if the consumer does not stop purchasing at $\alpha = \alpha_1$, the same must be true for all $\alpha > \alpha_1$ as well. This implies that the optimal profit under the blind box is weakly increasing in α.

In summary, we have proved that under bonus $v^+ > \check{v}$ and $\alpha \in [0, \alpha^{\max}]$:

1. The seller's profit under separate selling is independent of α.
2. The seller's profit under the blind box is weakly increasing in α.

To complete the first statement of the proposition, we consider two cases. If separate selling is more profitable at $\alpha = \alpha^{\max}$, set the threshold $\check{\alpha} = \alpha^{\max}$. Alternatively, if the blind box is more profitable at $\alpha = \alpha^{\max}$, set $\check{\alpha}$ to be the largest choice of $\alpha < \alpha^{\max}$ at which separate selling is at least as profitable as the blind box. It then follows that for all $\alpha < \check{\alpha}$, separate selling is the optimal mechanism.

For the sake of completeness, we prove that $\check{\alpha}$ can be strictly interior when $\alpha^{\max}$ is sufficiently large. Suppose that $\alpha^{\max} = 10n^2 H_n \cdot \frac{v^H}{v^L}$, where H_n denotes the n-th harmonic number, and $v^+ \geq \check{v}$ where $\check{v} > \frac{v^H(n+1+\alpha)-(n+1)v^L}{(1-\omega)^{n+1}}$. Consider $\alpha = \alpha^{\max}$, and set the blind box parameters as $\check{q} = \frac{1}{1+2nH_n}$ and $\check{p}_b = \frac{v^L \alpha/(2n)+v^+}{\bar{m}}$, where $\bar{m} \geq \{\frac{nH_n}{1-\check{q}}, \frac{1}{\check{q}}\}$ is the expected number of blind boxes purchased by consumers until they collect all $n+1$ unique items (this only depends on q).

We first argue that given these parameters, the consumer of type $(0,0)$ with the lowest valuation always purchases the blind box until they receive all $n+1$ items. We prove this by backward induction, showing that the expected value of purchasing is strictly positive at every state (n_r, n_s), where n_r and n_s denote the number of unique regular and special items collected, respectively.

There are two base cases. First, suppose that the consumer is in state $(n,0)$. The expected utility from purchasing the blind box until completion is given by

$$\begin{aligned}
U(n,0) &= v^L\left(1+\alpha(1-\frac{\check{q}n}{1-\check{q}})\right)+v^+-\check{p}_b\cdot\frac{1}{\check{q}}\\
&\geq v^L\left(1+\alpha\left(1-\frac{1}{2H_n}\right)\right)+v^+-\check{p}_b\cdot\frac{\bar{m}}{\frac{1}{\check{q}}}\cdot\frac{1}{\check{q}}\\
&= v^L+v^L\alpha-v^L\frac{\alpha}{2H_n}+v^+-v^L\frac{\alpha}{2n}-v^+\\
&= v^L+v^L\alpha\left(1-\frac{1}{2H_n}-\frac{1}{2n}\right)>0.
\end{aligned}$$

The inequality holds since $\bar{m}\geq\frac{1}{\check{q}}$. In the final step, we use the fact that $\frac{1}{2H_n}+\frac{1}{2n}\leq 1$, where $H_n\geq 1$ and $n\geq 1$. Next, consider the base case $(n-1,1)$. We have

$$\begin{aligned}
U(n-1,1) &= v^L+v^+-\check{p}_b\cdot\frac{n}{1-\check{q}}\\
&= v^L+v^+-\check{p}_b\left(\frac{1+2nH_n}{2H_n}\right)\\
&= v^L+v^+-\left(\frac{v^L\alpha/(2n)+v^+}{\bar{m}}\right)\left(\frac{1+2nH_n}{2H_n}\right)\\
&\geq v^L+v^+-\left(\frac{v^L\alpha/(2n)+v^+}{\bar{m}}\right)\frac{\bar{m}}{\frac{1}{\check{q}}}\left(\frac{1+2nH_n}{2H_n}\right)\\
&= v^L+v^+-\frac{v^L\alpha}{4nH_n}-\frac{v^+}{2H_n}\\
&= \frac{v^+}{2}+v^L-\frac{v^L\alpha}{4nH_n}>0.
\end{aligned}$$

The final inequality holds since $v^+>v^L\alpha$ by inequality (C.7) and $nH_n\geq 1$. A similar, though more involved argument holds for all other states (n_r,n_s) in the inductive case. First, consider the state $(n_r,1)$ where $n_r<n-1$. The consumer has the special item and needs to collect the $n-n_r$ missing regular items. We note that the expected number of purchases to achieve this goal is at most $\frac{nH_n}{1-q}$. We proceed as follows:

$$\begin{aligned}
U(n_r,1) &\geq (n-n_r)v^L+v^+-\check{p}_b\cdot\frac{1+2nH_n}{2nH_n}nH_n\\
&= (n-n_r)v^L+v^+-\left(\frac{v^L\alpha/(2n)+v^+}{2\bar{m}}\right)(1+2nH_n)\\
&\geq (n-n_r)v^L+v^+-\left(\frac{v^L\alpha/(2n)+v^+}{2}\right)\\
&= (n-n_r)v^L+\frac{v^+}{2}-\frac{v^L\alpha}{4n}>0.
\end{aligned}$$

The second inequality holds since $\bar{m}\geq 1/\check{q}=1+2nH_n$. The final inequality follows from the fact

that $v^+ > \frac{v^L \alpha}{2n}$. Next, consider the state $(n_r, 0)$. From this state, the consumer needs at most $\bar{m}$ blind boxes (in expectation) to complete the set. Therefore,

$$\begin{aligned} U(n_r,0) &\geq (n-n_r)v^L + v^L\left(1+\alpha\left(1-\frac{\check{q}n}{1-\check{q}}\right)\right) + v^+ - \check{p}_b \cdot \bar{m} \\ &= (n-n_r)v^L + v^L\left(1+\alpha\left(1-\frac{1}{2H_n}\right)\right) + v^+ - v^L\alpha/(2n) - v^+ \\ &\geq v^L + v^L\alpha\left(1-\frac{1}{2H_n}-\frac{1}{2n}\right) > 0. \end{aligned}$$

The final inequality holds since $n \geq 1$ and $H_n \geq 1$.

To conclude, since the consumer (type $(0,0)$) with the lowest valuations purchases all items until the set is completed, the same must be true for all other consumer types, as their valuations weakly dominate type $(0,0)$. Given this, the expected profit under this blind box policy at $\alpha^{\max}$ is given by

$$\begin{aligned} \check{\Pi}_B(\check{p}_b, \check{q}) &= (\check{p}_b - c)\cdot \bar{m} \\ &= \left(\frac{v^L\alpha^{\max}/(2n) + v^+}{\bar{m}} - c\right)\cdot \bar{m} \\ &= v^L\alpha^{\max}/(2n) + v^+ - c\cdot\bar{m} \\ &\geq v^+ + v^H\cdot 5nH_n - c(\frac{3}{2}+3nH_n) \\ &= v^+ + (\frac{3}{2}+3nH_n)(v^H - c) + v^H\cdot 2nH_n \\ &> v^+ + (n+1)(v^L - c) = \check{\Pi}_S. \end{aligned}$$

The first inequality holds by the definition of $\alpha^{\max} = 10n^2 H_n \frac{v^H}{v^L}$, and $\bar{m} \leq \frac{1}{\check{q}} + \frac{nH_n}{1-\check{q}} = \frac{3}{2} + 3nH_n$. The optimal profit under separate selling from Equation (C.6) is given by $\check{\Pi}_S = (n+1)(v^L - c) + v^+$. And the final inequality always holds since $v^H - c \geq v^L - c$ and $(\frac{3}{2} + 3nH_n) > n+1$. Thus, we have constructed a set of parameters such that the blind box is strictly more profitable at $\alpha = \alpha^{\max}$. Since the profit under blind boxes is weakly increasing in α, there must exist some threshold $\check{\alpha} < \alpha^{\max}$ such that separate selling is strictly profitable for values of α smaller than $\check{\alpha}$. This completes the proof of the first statement.

Proof of Statement (ii).

Recall that when the collection bonus is sufficiently large, separate selling outperforms blind box if and only if the status parameter α is smaller than some threshold $\check{\alpha}$. In this part of the proof, we first derive a lower bound on the threshold $\check{\alpha}$ as a function of c and n. Following this, we derive a similar bound on α^H, which is the smallest threshold at which the blind box is optimal in the

absence of the bonus. By comparing $\check{\alpha}$ with α^H, we identify conditions under which the blind box is optimal in the absence of the bonus, but the converse holds under a sufficiently large bonus. This yields the second statement of the proposition.

Suppose that we are given a blind box with probability q of drawing the special item, and let $\bar{m}$ denote the expected number of blind boxes purchased by consumers to obtain all $n+1$ unique items (notably, $\bar{m}$ is independent of consumer type). For convenience, define $V^L_{\text{total}} = nv^L + v_s(q, v^L) + v^+$, where $v_s(q, v^L) = v^L\left(1 + \alpha\left(1 - \frac{qn}{1-q}\right)\right)$ is the valuation for the special item for consumer type $(0,0)$. We begin by proving that when the bonus is sufficiently large, under the blind box, any price $p_b > \frac{V^L_{\text{total}}}{\bar{m}}$ is strictly suboptimal for any given q.

Specifically, we prove by contradiction that the optimal blind box price must satisfy $p^*_b \leq \frac{V^L_{\text{total}}}{\bar{m}}$ for any given $q \in (0, \frac{1}{n+1}]$. Suppose, for contradiction, that the optimal price satisfies $p^*_b > \frac{V^L_{\text{total}}}{\bar{m}}$ for a given q. From Lemma E7, under this price, the fraction $\bar{f}_B$ of consumers that do not collect the bonus satisfies $\bar{f}_B \geq (1-\omega)^{n+1} \cdot \frac{1}{n+1}$. Then, we derive an upper bound on the seller's profit as follows:

$$\check{\Pi}_B(p^*_b, q) \leq (1 - \bar{f}_B)\left[nv^H + v^H(1+\alpha) + v^+ - c\bar{m}\right] + \bar{f}_B \cdot v^H(n+1+\alpha), \tag{C.8}$$

In the above expression, we use the fact that the total expected revenue from any consumer type is at most their ex-ante valuation for the items, which is bounded from above by $nv^H + v^H(1+\alpha)$. Additionally, consumers who collect all items receive the bonus v^+. We ignore costs for consumers who do not collect the bonus. Now, consider an alternative pricing policy where all consumers receive the bonus. Specifically, suppose that $\check{p}_b = \frac{v^+}{\check{m}}$ for $\check{q} = \frac{1}{n+1}$, where $\check{m}$ denotes the expected number of purchases when the consumer collects all $n+1$ items under $\check{q}$. We claim that at this price, all consumer types, including the lowest-value type $(0,0)$, will continue purchasing the blind box until they collect all $n+1$ items. To see why, note that the expected utility for consumer type $(0,0)$ from purchasing until completion given any state (n_r, n_s) is at least $(n - n_r)v^L + (1 - n_s)v_s(\check{q}, v^L) + v^+ - \check{m} \cdot \frac{v^+}{\check{m}} > 0$. Since all other consumer types dominate $(0,0)$ in terms of their stopping point, it follows that every consumer will purchase the blind box repeatedly to complete the collection when $p_b = \frac{v^+}{\check{m}}$. The seller's expected profit is

$$\check{\Pi}_B(\check{p}_b, \check{q}) = \check{m}\left(\frac{v^+}{\check{m}} - c\right) = v^+ - c\check{m}. \tag{C.9}$$

We now show that the alternative policy yields strictly higher profit when $v^+ > \check{v}$. To do so, we use the fact that $\bar{m} \geq \check{m}$ since the expected number of purchases to collect all items is weakly

decreasing in $q \in (0, \frac{1}{n+1}]$. Comparing Equations (C.8) and (C.9), the alternative policy dominates if

$$v^+ - c\check{m} > v^H(n+1+\alpha) + (1-f_B)(v^+ - c\bar{m}) \implies v^+ > c\check{m} + \frac{v^H(n+1+\alpha)}{f_B}.$$

Moreover, when $\check{q} = \frac{1}{n+1}$, we have $\check{m} = (n+1)H_{n+1}$ from the coupon collector problem. Using the bound $f_B \geq (1-\omega)^{n+1} \cdot \frac{1}{n+1}$ from Lemma E7 and the fact that $c \leq v^H$, we get that the alternative policy is more profitable as long as

$$v^+ > v^H \left((n+1)H_{n+1} + \frac{(n+1+\alpha^{\max})(n+1)}{(1-\omega)^{n+1}} \right).$$

Given the assumption $v^+ > \check{v}$ and the definition of $\check{v}$, we infer that $\check{\Pi}_B(\check{p}_b, \check{q}) > \check{\Pi}_B(p_b^*, q)$. This contradicts the optimality of $p_b^* > \frac{V_{\text{total}}^L}{\bar{m}}$. Therefore, we conclude that $p_b^* \leq \frac{V_{\text{total}}^L}{\bar{m}}$ for any given q. Having established that the optimal blind box price satisfies $p_b^* \leq \frac{V_{\text{total}}^L}{\bar{m}}$, we now characterize an upper bound on the profit under blind box. For any given $p_b \leq \frac{V_{\text{total}}^L}{\bar{m}}$ and q, the profit under blind box is $\check{\Pi}_B \leq m\left(\frac{V_{\text{total}}^L}{\bar{m}} - c\right) = V_{\text{total}}^L - c\bar{m} = nv^L + v_s(q, v^L) + v^+ - c\bar{m}$.

To compare the above profit to separate selling, we first set a lower bound on $\bar{m}$, the expected number of purchases of the blind box required to collect all items. In particular, m satisfies $\bar{m} \geq \max\left(\frac{1}{q}, (n+1)H_{n+1}\right)$. The first term $\frac{1}{q}$ represents the expected number of purchases needed to collect the special item whereas the second term $(n+1)H_{n+1}$ is the expected number of purchases in the classical coupon collector problem when all items are equally likely with $q = \frac{1}{n+1}$. Using this lower bound, we can write

$$\check{\Pi}_B(q) \leq nv^L + v^L(1 + \alpha(1 - \frac{qn}{1-q})) + v^+ - c \cdot \min\left(\frac{1}{q}, (n+1)H_{n+1}\right) = y(q). \tag{C.10}$$

From Statement (i), Equation (C.6), the optimal profit under separate selling when $v^+ > \check{v}$ is $\check{\Pi}_S^* = (n+1)(v^L - c) + v^+$. Suppose that $K = (n+1)H_{n+1}$. Then, we claim that $\check{\Pi}_B^* \leq \check{\Pi}_S^*$ as long as $\alpha \leq \check{\alpha} = \frac{c(K-1)}{v^L}$. We break this down in the following steps:

- Consider $y(q)$ in Equation (C.10). If $q < \frac{1}{K}$, $\frac{\partial y}{\partial q} = -\frac{v^L \alpha n}{(1-q)^2} < 0$ for any given α. Thus, the optimal q satisfies $q^* \geq \frac{1}{K}$. Moreover, $\frac{\partial y}{\partial q} = -\frac{v^L \alpha n}{(1-q)^2} + \frac{c}{q^2}$ for $q \geq \frac{1}{K}$. When $\alpha \leq \frac{c}{v^L n}(K-1)^2$, we have $q^* = \frac{1}{K}$. Since $\check{\alpha} \leq \frac{c}{v^L n}(K-1)^2$, this implies that $q^* = \frac{1}{K}$ holds for all $\alpha \leq \check{\alpha}$.
- Substituting $q^* = \frac{1}{K}$ into the blind box profit, we get

$$y(q^*) = (n+1)v^L - Kc + v^+ + v^L \alpha \left(\frac{K-(n+1)}{K-1} \right).$$

- Comparing the above profit to separate selling, we get that $\check{\Pi}_S^* \geq \check{\Pi}_B^*(q^*)$ as long as

$$(n+1)v^L - (n+1)c + v^+ \geq (n+1)v^L - Kc + v^+ + v^L\alpha\left(\frac{K-(n+1)}{K-1}\right).$$

  The above expression implies that $c(K-(n+1)) \geq v^L\alpha\left(\frac{K-(n+1)}{K-1}\right)$. This is equivalent to $\alpha \leq \frac{c}{v^L}(K-1) = \check{\alpha}$.

In conclusion, we have proved that when $\alpha \leq \check{\alpha} = \frac{c}{v^L}(K-1)$, it must be the case that separate selling outperforms blind box as long as $v^+ > \check{v}$. To conclude the proof, we now establish the second statement of the proposition by comparing $\check{\alpha}$ to the corresponding threshold α^L in the absence of the bonus.

We define the constants $\check{\omega}$ and $\check{n}$ from the proposition statement as they become pertinent. First, in the absence of the bonus, we have the following lower bound for profit under blind box obtained by setting $p_b = q_H v_s(q_H, v^H)$ where $q_H = \frac{1}{1+\sqrt{n\alpha v^H/c}}$ (see Lemma E2):

$$\Pi_B^* \geq \omega\left((1+\alpha)v^H - c - 2\sqrt{n\alpha v^H c}\right). \tag{C.11}$$

In a similar fashion, we now establish an upper bound on the profit under separate selling, in the absence of the bonus. Specifically, when $c \geq v^L$, we follow a similar approach as in the proof of Proposition 1 (e.g., see Equation (B.1)), and get that

$$\Pi_S^* \leq n(v^H - c) + \max_{x_s \in [0,1]} x_s(v_0(1+\alpha(1-x_s)) - c), \tag{C.12}$$

where $v_0 = v^H$ if $x_s \leq \omega$ and $v_0 = v^L$ if $x_s > \omega$. Define $\check{\omega} := \frac{v^H(1+\alpha)-c}{2\alpha v^H}$. Optimizing the expression for Π_S^* over $x_s \in [0,1]$, we get that as long as $\omega \geq \check{\omega}$, the maximum of the right-hand side of (C.12) is given by $x_s^* = \frac{v^H(1+\alpha)-c}{2\alpha v^H}$. In this case, we get the following upper bound on Π_S^*:

$$\Pi_S^* = n(v^H - c) + \frac{(v^H(1+\alpha)-c)^2}{4\alpha v^H}. \tag{C.13}$$

Combining Equations (C.11) and (C.13), blind box dominates as long as

$$\omega((1+\alpha)v^H - c - 2\sqrt{n\alpha v^H c}) \geq n(v^H - c) + \frac{(v^H(1+\alpha)-c)^2}{4\alpha v^H}.$$

We now establish conditions under which the above inequality strictly holds for all $\alpha \geq \check{\alpha}$. This in turn implies that the cutoff α^L above which blind box dominates (in the absence of the bonus) is smaller than $\check{\alpha}$. The proposition then follows.

Suppose that $D(\alpha, n) = \Pi_B^*(\alpha) - \Pi_S^*(\alpha)$. This term can be written as

$$D(\alpha, n) = \omega v^H \alpha + \omega(v^H - c) - 2\omega\sqrt{n\alpha v^H c} - n(v^H - c)$$

$$-\frac{(v^H-c)^2}{4\alpha v^H}-\frac{v^H-c}{2}-\frac{v^H\alpha}{4}.$$

First, we establish the existence of a threshold $\hat{n}$ so that for all $n>\check{n}$, $D(\alpha,n)$ is increasing in α. Following this, we show that $D(\alpha,n)>0$ at $\alpha=\check{\alpha}$ for all $n>\check{n}$. Towards the first goal, we have

$$\frac{\partial D}{\partial\alpha}=v^H\left(\omega-\frac{1}{4}\right)-\omega\sqrt{nv^Hc}\cdot\alpha^{-1/2}+\frac{(v^H-c)^2}{4v^H}\cdot\alpha^{-2}.$$

For $\alpha\geq\check{\alpha}$, the (absolute value of the) second term is at most to $\omega\sqrt{v^Hv^L}\cdot\sqrt{\frac{n}{(n+1)H_{n+1}-1}}\leq\omega v^H\sqrt{\frac{1}{H_{n+1}-1}}$. Suppose that $\check{n}_1$ denotes the smallest value of n at which $H_{n+1}-1=4$. Then it is easy to conclude that for all $n\geq\check{n}_1$ and $\alpha\geq\check{\alpha}$, $\frac{\partial D}{\partial\alpha}>0$. The expression is strictly positive since $\omega>\frac{1}{2}$. Next, we show the existence of a constant $\check{n}_2>0$ such that $D(\check{\alpha},n)\geq 0$ for $n\geq\check{n}_2$. Substituting $\alpha=\check{\alpha}(n)$ into the expression for $D(\alpha,n)$, we have

$$\begin{aligned}D(\check{\alpha},n)&=v^H\left(\omega-\frac{1}{4}\right)\check{\alpha}-\omega(v^H-c)\left(\frac{n}{\omega}-1+\frac{1}{2\omega}\right)-2\omega\sqrt{n\check{\alpha}v^Hc}-\frac{(v^H-c)^2}{4\check{\alpha}v^H}\\&\geq v^H\left(\omega-\frac{1}{4}\right)\check{\alpha}-n(v^H-c)-2\omega\sqrt{n\check{\alpha}v^Hc}-\frac{(v^H-c)^2}{4\check{\alpha}v^H}\\&\geq\sqrt{n\check{\alpha}}\left[v^H\left(\omega-\frac{1}{4}\right)\sqrt{\frac{\check{\alpha}}{n}}-(v^H-c)\sqrt{\frac{n}{\check{\alpha}}}-2\omega\sqrt{v^Hc}-\frac{(v^H-c)^2}{4\check{\alpha}\sqrt{n\check{\alpha}}.v^H}\right].\end{aligned}$$

The first inequality holds since $\omega>1/2$ and the second inequality comes from the fact that $\check{\alpha}/n=\frac{c}{v^L}(H_n-\frac{1}{n})$ increases with n. Then, the term inside the square parentheses increases with n. There must exist a cutoff $\check{n}_2$ at which $D(\check{\alpha},\check{n}_2)>0$. Define the overall cutoff $\check{n}=\max(\check{n}_1,\check{n}_2)$.

We are now ready to complete the proof of Statement (ii). For all $n\geq\check{n}$ and $\alpha\geq\check{\alpha}$, we have $\Pi_B^*>\Pi_S^*$ in the absence of the bonus. This implies that for any $n\geq\check{n}$, the threshold α^L above which blind boxes are optimal is strictly smaller than $\check{\alpha}$. This completes the proof. □

D. Extensions

D.1. Salvage Value for Repeat Items

Our base model in Section 2 assumed that consumers only derive value from the first copy of any given item, and that all duplicate copies yield no additional utility. However, while additional copies of items often yield less marginal utility than the first, consumers may still place some value on them. This could occur, for example, when consumers can resell additional copies on the secondary market at a discounted price. The incorporation of a discount for additional copies could fundamentally alter the dynamics of both selling mechanisms; e.g., sellers could significantly lower the price of individual items or the blind box itself to encourage repeat purchases. In this section,

we generalize our base model to allow for reduced values for duplicates and study how this affects the seller's decisions.

For simplicity, we consider a special case of the base model in which $v^L = v^H = v$, i.e., consumers are homogeneous in their valuation of regular items. Similarly, all consumers have the same base reward $v_0 = v$ for the special item in the absence of rarity. We capture diminishing but non-zero utility for repeat consumption by assuming that the consumer's value for the h-th copy of item $i \in [n]$ is given by $\gamma^{h-1}v$, where $\gamma \in [0,1)$ is a discount factor that reflects the decreasing marginal utility from repeated consumption. In a similar vein, the base reward for the h-th copy of the special item is also $\gamma^{h-1}v$. Note that when $\gamma = 0$, this reduces to our base model (with $v^L = v^H = v$) in Section 2. We characterize both selling mechanisms in the presence of a salvage value below.

Separate Selling. Under separate selling, the seller can set price $p_r = \gamma^{h-1}v$ for any $h \in \{0,1,2,\ldots\}$ for the regular items. For consistency with our base model and tractability, we restrict our focus to outcomes in which the seller sets the same price for all regular items. Given price $p_r = \gamma^{h-1}v$, consumers proceed to purchase the same item h times. For the special item, the seller can choose $p_s = (1+(1-r)\alpha)\, v\gamma^{\lceil x_s \rceil - 1}$ for $r = x_s/h$ and $x_s \in [0,h]$. For instance, when $x_s = 1.5$, a fraction of consumers purchase the special item twice, which occurs at price $p_s = (1+(1-r)\alpha)\, v\gamma$. Note that by selling $h \geq 1$ copies of the regular items, the seller is able to boost the relative rarity of the special item. The seller's problem is given by

$$\max_{x_s,h} \Pi_S(\gamma) = nh(\gamma^{h-1}v - c) + x_s((1+(1-r)\alpha)\, v\gamma^{\lceil x_s \rceil - 1} - c).$$

Blind Box. Let $\boldsymbol{h} = (h_0, h_1, \ldots, h_n)$ be an $(n+1)$-dimensional vector where h_i denotes the number of copies of the i-th regular item possessed by the consumer and h_0 is the number of copies of the special item. Then, given blind box parameters (p_b, q), the consumer's expected utility from purchasing the blind box after collecting $\mathbf{h}$ items is given by $U(\boldsymbol{h}) = \sum_{i=1}^{n} \frac{(1-q)}{n}\gamma^{h_i}v + \gamma^{h_0}qv(1+(1-\frac{qn}{1-q})\alpha) - p_b$. The consumer purchases the blind box as long as $U(\boldsymbol{h}) \geq 0$. Once again, the seller's decision is given by

$$\max_{p_b,q} \Pi_B(\gamma) = E[m](p_b - c),$$

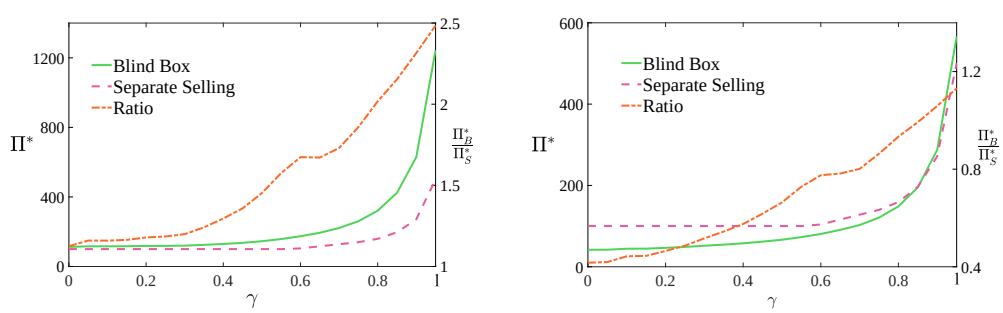
where m denotes the number of blind boxes purchased before the consumer reaches their stopping condition.

Our main result in this section demonstrates that as γ - the discount factor corresponding to repeat consumptions - increases, the performance of the blind box relative to separate selling grows as long as the status parameter α is bounded.

Lemma D1. *There exists a threshold α_g such that for all $\alpha < \alpha_\gamma$, the ratio $\Pi_B^*(\gamma)/\Pi_S^*(\gamma)$ is increasing in $\gamma \in [0, \epsilon(\alpha)]$ up to some suitable discount level $\epsilon(\alpha)$.*

The proof of the lemma is deferred to Appendix D.3. Lemma D1 shows that as the discount factor γ increases from the base case of $\gamma = 0$, the blind box becomes increasingly profitable relative to separate selling. This disparity in performance stems from how the value for repeat consumption impacts the seller's pricing policy and consumers' stopping points. Specifically, when γ is small, under separate selling, the seller must reduce prices substantially to encourage additional purchases, which may not lead to higher profits. For example, when $\gamma < 1/2$, the price of any given regular item must be reduced by a factor (of at least) $\frac{1}{2}$ to induce consumers to make a second purchase. Although repeat purchases can increase the rarity of the special item, the additional profit from it may not offset the loss from regular items. This is particularly true when consumers are not strongly status-seeking. In contrast, under the blind box mechanism, even if the seller utilizes the same price as the base model ($\gamma = 0$), any increase in γ can only delay the stopping condition, leading to additional purchases. This is a consequence of the fact that the consumer's expected marginal value at any given state increases with γ as duplicate copies add to the total value.

Figure D.1 depicts the optimal profits under both schemes, as well as the ratio $\Pi_B^*(\gamma)/\Pi_S^*(\gamma)$ across different values of γ. As γ increases, the optimal profit under blind box rises, while that under separate selling remains flat for small γ and increases thereafter. The upward trend of $\Pi_B^*(\gamma)/\Pi_S^*(\gamma)$ indicates that incorporating salvage value for repeat purchases makes the blind box more advantageous.



(a) $c/v = 0.4$ (b) $c/v = 0.8$

Figure D.1 Optimal Profit Under Blind Box and Separate Selling at Different Values of $\gamma \in [0, 1]$.

Notes. Parameters: $n = 12, v = 10, \alpha = 40$. For any fixed value of c/v and α, the profits under both schemes increase with γ. However, the blind box increases at a faster rate when γ is small.

D.2. Two-Stage Newsvendor

In our base model, we assumed that the seller can produce the exact quantity of items purchased by consumers under the blind-box model. Therefore, the seller does not incur any unnecessary costs for unsold items, nor face any lost sales due to supply shortages. In this section, we extend this model by assuming that the seller, in addition to choosing prices and rarity, also determines the optimal production quantity in advance of the selling. Moreover, we assume that the uncertainty in sales impacts the blind box mechanism but not the separate selling. We note that under the blind box mechanism, there are two sources of uncertainty: the different sample paths followed by each individual consumer and the consumer's random valuations. However, under separate selling, the consumer's purchase decisions are deterministic once valuations are known. As such, our model in this section seeks to spotlight the production inefficiencies stemming from the randomness in sample paths (and hence items sold) under the blind box model.

We incorporate the production decision as a two-stage newsvendor problem on top of the base model. Specifically, the seller chooses a production quantity Q before the sales season, incurring a unit cost c. If the realized blind box demand m exceeds Q, the seller may place a second-stage order at unit cost $c_H > c$ to fill the shortfall. There is no salvage value for leftovers. Given this model, the decision timeline works as follows:

1. The seller selects the optimal price p_b^* and probability q^* of drawing the special item that maximizes the expected profit in our base model.

$$\max_{p_b, q} \Pi_B = (p_b - c) \cdot E[m].$$

2. Given p_b^* and q^*, we obtain the empirical sales distribution $m \sim F$.
3. The optimal production quantity Q^* is determined as follows. Based on the marginal underage and overage costs of $C_u = c_H - c$ and $C_o = c$, respectively, we determine the optimal production quantity Q^* by solving

$$F(Q^*) = \frac{C_u}{C_u + C_o} = \frac{c_H - c}{c_H}.$$

4. Finally, we calculate the realized profit as

$$\Pi_B^* = p_b \, \mathbb{E}[m] - cQ^* - c_H \, \mathbb{E}\big[(m - Q^*)^+\big].$$

If emergency replenishment is not profitable, i.e., $p_b^* < c_H$, the problem reduces to the one-shot newsvendor:

$$\Pi_B^* = p_b^* \, \mathbb{E}\big[\min(Q^*, m)\big] - cQ^*, \quad \text{where } F(Q^*) = \frac{p_b^* - c}{p_b^*}.$$

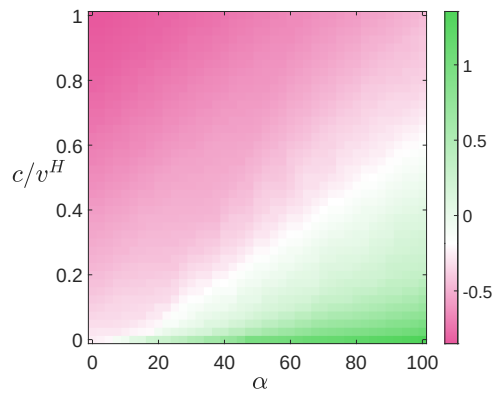


Figure D.2 The Ratio of $(\Pi_B^* - \Pi_S^*)/\Pi_S^*$ in the Two-stage Newsvendor Setting at Different Values of α and c/v^H.

Notes. Parameters: $n = 12, v^H = 10, v^L = 5, \omega = 0.8, c_H = 10$.

In Figure D.2, we compare the two-stage newsvendor profit under blind box with the deterministic profit under separate selling (i.e., the profit under separate selling matches that in Section 3).

Compared with Figure 3(a), the region where separate selling dominates becomes larger. This is to be expected, as the blind box becomes less favorable when the seller must make irrevocable decisions on production quantity in advance. As mentioned previously, this stems from the randomness in the number of blind boxes purchased by consumers, which in turn depends on the sequence of draws. However, blind boxes still outperform separate selling when costs are low or when consumers exhibit strong status-seeking preferences. Therefore, our core findings remain robust even with the incorporation of *ex-ante* production decisions.

D.3. Proofs from Appendix D: Extensions

Proof of Lemma D1 To prove the lemma, we show that Π_B^* weakly increases with γ, while Π_S^* is non-increasing in γ for $\gamma \in [0, \epsilon(\alpha)]$ for $\alpha < \alpha_g$, for a suitable threshold α_g. Consequently, for $\alpha < \alpha_g$, the ratio Π_B^*/Π_S^* is weakly increasing in γ for $\gamma \in [0, \epsilon(\alpha)]$.

Blind Box. We first show that Π_B^* increases with γ in the above range. To prove this, we show that the consumer's stopping state is delayed further and further as γ increases. Formally, given any blind box (p_b, q), suppose $\boldsymbol{h}$ denotes the consumer's state where $h_i \geq 0$ is the number of copies of regular item $i \in [n]$ collected, and h_0 is the number of copies of the special item. Then the consumer continues purchasing as long as

$$U(\boldsymbol{h}) = \sum_{i=1}^{n} \frac{(1-q)}{n} \gamma^{h_i} v + \gamma^{h_0} q v (1 + (1 - \frac{qn}{1-q})\alpha) - p_b \geq 0.$$

Crucially, we notice that $U(\boldsymbol{h})$ is strictly increasing in γ. Therefore, suppose that $h(\gamma)$ denotes any state where $U(\boldsymbol{h}(\gamma)) \geq 0$ for a given $\gamma > 0$. Then, for any $\gamma' > \gamma$, $U(\boldsymbol{h}(\gamma)) \geq 0$ still holds. In other words, if the consumer does not stop purchasing at a given $\boldsymbol{h}, \gamma$, then this is also true for all $\gamma' > \gamma$ at the same state $\boldsymbol{h}$. The expected number of blind boxes purchased by consumers is weakly increasing in γ for all (p_b, q). In conclusion, since the seller's profit is weakly increasing in γ for all (p_b, q), this must be true for the optimal profit Π_B^* as well. Notably, this is true for all α, γ

Separate Selling. We next show that Π_S^* is non-increasing in γ when both γ and α are sufficiently small. Recall that the profit under separate selling is given by

$$\Pi_S = nh\left(v\gamma^{h-1} - c\right) + x_s\left(\left(1 + \left(1 - \frac{x_s}{h}\right)\alpha\right) v\gamma^{\lceil x_s \rceil - 1} - c\right).$$

We begin by proving (by contradiction) that under the separate selling, the profit obtained under any $x_s > 1$ is strictly dominated by one under $x_s' \in [0,1]$ when γ is sufficiently small. First, consider the case where $h = 1$. In this case, we know $x_s \leq h \in [0,1]$, and so the claim holds trivially. Next, consider a pricing outcome where $h > 1$. We argue that the seller's profit for any $x_s > 1$ is strictly dominated by the profit under an alternative solution with the same prices for regular items but where $x_s' = 1$. Given $p_r = v\gamma^{h-1}$, the seller's profit can be written as

$$\Pi_S = nh\left(v\gamma^{h-1} - c\right) + x_s\left(\left(1 + \left(1 - \frac{x_s}{h}\right)\alpha\right) v\gamma^{\lceil x_s \rceil - 1} - c\right).$$

Under an alternative policy where the regular items are still priced at $p_r = v\gamma^{h-1}$, but the special item now has $x_s' = 1$, the seller's profit can be written as

$$\Pi_S' = nh\left(v\gamma^{h-1} - c\right) + \left(\left(1 + \left(1 - \frac{1}{h}\right)\alpha\right) v - c\right).$$

Note that at $\gamma = 0$, we have

$$\begin{aligned}
\Pi_S &= -nhc - cx_s \\
&< -nhc + \left(\left(1 + \left(1 - \frac{1}{h}\right)\alpha\right) v - c\right) \\
&= \Pi_S'.
\end{aligned}$$

The second inequality holds since $v \geq c$, $x_s > 1$ and $\frac{1}{h} < 1$. The inequality is strict since $\alpha > 0$. Therefore, when $\gamma = 0$, it is strictly more profitable for the seller to choose $x_s' \leq 1$ for any given $h \geq 1$. Consequently, for any given $\alpha > 0$, there must exist a cutoff γ' such that for all $\gamma \in [0, \gamma']$, we have $x_s^* \in [0,1]$ for the optimal solution under separate selling.

Having proved that $x_S^* \in [0,1]$ for $\gamma \in [0, \gamma']$, we now establish that when α is sufficiently small, the optimal number of regular items per consumer is $h^* = 1$. For a given $p_r^*(h) = v\gamma^{h-1}$, each consumer

purchases exactly h copies of every regular item. For now, assume that $x_s \in [0,1]$. The profit under separate selling is $\Pi_S(p_r^*(h), x_s) = nh\left(v\gamma^{h-1} - c\right) + x_s\left(\left(1 + \left(1 - \frac{x_s}{h}\right)\alpha\right)v - c\right)$. Optimizing over $x_s \in [0,1]$, we get

$$x_s^*(h) = \min\left\{1, \max\left\{0, \frac{h\left((1+\alpha)v - c\right)}{2\alpha v}\right\}\right\}.$$

Consider the family of strategies given by $(p_r(h), x_s^*(h))_{h\geq 1}$ and suppose that $\alpha < \alpha_g := n(\frac{v-c}{v})$. For any $h > 1$, we show that profit under $(p_r(h), x_s^*(h))$ is strictly dominated by $(p_r(1), x_s^*(1))$ when α is sufficiently small. We can bound the difference in profits as follows:

$$\begin{aligned}\Pi_S(p_r^*(1), x_s^*(1)) - \Pi_S(p_r^*(h), x_s^*(h)) &\geq \Pi_S(p_r^*(1), x_s^*(h)) - \Pi_S(p_r^*(h), x_s^*(h)) \\ &= n\left(v - c\right) - nh\left(v\gamma^{h-1} - c\right) - x_s^2 v\alpha\left(1 - \frac{1}{h}\right).\end{aligned}$$

In the first step, we use the fact that when the price is $p_r^*(1)$, choosing $x_s = x_s^*(h)$ is suboptimal. However, this allows for tractability. We now consider two cases. First, suppose that $v\gamma^{h-1} - c \leq 0$. In this scenario, the proof is straightforward:

$$\begin{aligned}\Pi_S(p_r^*(1), x_s^*(1)) - \Pi_S(p_r^*(h), x_s^*(h)) &\geq n\left(v - c\right) + nh\left(c - v\gamma^{h-1}\right) - x_s^2 v\alpha\left(1 - \frac{1}{h}\right) \\ &\geq n\left(v - c\right) - v\alpha > 0.\end{aligned}$$

The final step holds as long as $\alpha < \alpha_g = n(1 - \frac{c}{v})$. Next, suppose that $v\gamma^{h-1} - c > 0$. In this scenario, we assume $\gamma \leq \frac{1}{2}$ and make use of the fact $h \cdot \gamma^{h-1} \leq 2\gamma$ for all $h \geq 2$. This gives us

$$\begin{aligned}\Pi_S(p_r^*(1), x_s^*(1)) - \Pi_S(p_r^*(h), x_s^*(h)) &\geq n\left(v - c\right) - nh\left(v\gamma^{h-1} - c\right) - x_s^2 v\alpha\left(1 - \frac{1}{h}\right) \\ &\geq n\left(v + c - 2v\gamma\right) - v\alpha \\ &> 0.\end{aligned}$$

The final expression holds for all $\alpha < \frac{n(v+c-2v\gamma)}{v}$. To conclude, we have just proved that for any given $\gamma \leq \frac{1}{2}$ and $\alpha \leq \min(\alpha_g, \frac{n(v+c-2v\gamma)}{v})$, the optimal policy under separate selling with $x_s \in [0,1]$ involves $p_r^* = v$. i.e., consumers purchase exactly one unit of each regular item. In order to combine the two parts of the proof, for any given $\alpha < \alpha_g = n(\frac{v-c}{v})$, define $\epsilon(\alpha)$ to be the largest value of γ up to which the previous conditions hold, namely,

$$\epsilon(\alpha) = \min(\gamma', \frac{1}{2}, \frac{1}{2} + \frac{c}{2v} - \frac{\alpha}{2n}).$$

Then by the same argument made earlier, for any $\alpha < \alpha_g$ and $\gamma \leq \epsilon(\alpha)$, the optimal policy under separate selling involves $x_s^* \in [0,1]$ and $p_r^* = v$. Consequently, $h^* = 1$ under separate selling. In this case, the profit under separate selling is independent of γ. Since Π_S^* does not depend on γ, it is trivially non-increasing in $\gamma \in [0, \epsilon(\alpha)]$ when $\alpha < \alpha_g$.

Given that Π_B^* is weakly increasing in γ, and Π_S^* is constant in γ, it follows that the ratio Π_B^*/Π_S^* is weakly increasing in $\gamma \in [0, \epsilon(\alpha)]$ for $\alpha < \alpha_g$. □

E. Auxiliary Results

E.1. Properties of the Blind Box and Separate Selling

Under separate selling, the price of regular items is $p_r(i) \in \{v^L, v^H\}$. The following lemma shows that the number of unique regular items for which $p_r(i) = v^L$ is increasing in α, i.e., as consumers' value for rarity increases, the seller prefers to underprice more regular items to boost its sales and consequently, amplify the rarity of the special item.

Lemma E1. *For any given α, suppose that $k^*(\alpha)$ denotes the number of regular items for which the optimal price under separate selling is $p_r(i) = v^L$. Then:*

1. *$k^*(\alpha)$ is non-decreasing in α.*
2. *There exists a threshold α_n such that for all $\alpha \geq \alpha_n$, we have $k^*(\alpha) = n$.*

Proof of Lemma E1. The seller's profit can be expressed as a function of k (the number of regular items priced at $p_r(i) = v^L$) and x_s (the supply of the special item) as follows:

$$\Pi_S(k, x_s; \alpha) = k(v^L - c) + (n-k)\omega(v^H - c) + x_s \left[(1+\alpha)v_0 - c - \alpha v_0 \frac{n x_s}{k + (n-k)\omega}\right],$$

where $v_0 = v^H$ if $x_s \leq \omega$ and $v_0 = v^L$ if $x_s > \omega$ as defined in Section 2.

We begin by proving that $k^*(\alpha)$ is non-decreasing in α. Towards this end, the cross-partial derivative of Π_S with respect to k and α for any given x_s (and therefore, v_0) is given by

$$\frac{\partial^2 \Pi_S}{\partial \alpha \partial k} = \frac{\partial}{\partial \alpha}\left[\frac{\partial}{\partial k}\left(-\frac{\alpha v_0 n x_s^2}{k(1-\omega) + n\omega}\right)\right] = \frac{v_0 n x_s^2 (1-\omega)}{[k + (n-k)\omega]^2}.$$

Since this expression is non-negative for both $v_0 = v^H$ and $v_0 = v^L$, the profit function $\Pi_S(k, x_s; \alpha)$ has increasing differences in (k, α) for any fixed x_s. Consequently, by the aggregation principle (see Milgrom and Shannon (1994), Corollary 3), it follows that the above property also holds at the optimal $x_s \in [0,1]$ for any given k, α. Formally, let $\Pi_S^*(k, \alpha) = \max_{x_s \in [0,1]} \Pi_S(k, x_s; \alpha)$. Then, this function also satisfies

$$\frac{\partial^2 \Pi_S^*}{\partial \alpha \partial k} \geq 0.$$

With $\Pi_S^*(k, \alpha)$ established as having increasing differences in (k, α), we can apply Theorem 5 from Milgrom and Shannon (1994). By this theorem, if $\Pi_S^*(k, \alpha)$ has increasing differences in k (the number of items priced at $p_r(i) = v^L$) and α, then the optimal number of low-priced regular items $k^*(\alpha) = \arg\max_k \Pi_S^*(k, \alpha)$, is non-decreasing with respect to α. Intuitively, this holds because at higher values of α, the seller is incentivized to increase the supply of regular items (by lowering prices), which boosts the rarity of the special item and its value.

Part 2: Existence of threshold α_n.

We analyze the behavior of $k^*(\alpha)$ at extreme values of α.

(i) At $\alpha = 0$. The profit function becomes $\Pi_S(k, x_s; 0) = k(v^L - c) + (n-k)\omega(v^H - c) + x_s(v_0 - c)$. In this regime, the choice of k does not affect the profit from the special item. Therefore, we have $k^*(0) = 0$ if $\omega(v^H - c) > v^L - c$, and $k^*(0) = n$ if $\omega(v^H - c) \leq v^L - c$.

(ii) As $\alpha \to \infty$. Consider the optimal profit normalized by α:

$$\frac{\Pi_S^*(k,\alpha)}{\alpha} = \max_{x_s \in [0,1]} \left\{ \frac{k(v^L - c) + (n-k)\omega(v^H - c) + x_s(v_0(x_s) - c)}{\alpha} + x_s v_0(x_s) \left(1 - \frac{n x_s}{k + (n-k)\omega}\right) \right\},$$

As $\alpha \to \infty$, the first term in the parentheses vanishes. Therefore, the limit is given by:

$$B(k) \equiv \lim_{\alpha \to \infty} \frac{\Pi_S^*(k,\alpha)}{\alpha} = \max_{x_s \in [0,1]} \left\{ x_s v_0(x_s) \left(1 - \frac{n x_s}{k + (n-k)\omega}\right) \right\}.$$

It is straightforward to see that for any given $x_s \in (0,1]$, the term inside the parentheses above is maximized at the maximum value of $k(1-\omega) + n\omega$ which occurs when $k = n$. By the first part of this lemma, it follows that for any specific $k \in \{0, \ldots, n-1\}$, there must exist a threshold $\hat{\alpha}_k$ such that for all $\alpha > \hat{\alpha}_k$, we have $\Pi_S^*(n,\alpha) > \Pi_S^*(k,\alpha)$. By defining $\alpha_n = \max_{k \in \{0,\ldots,n-1\}} \{\hat{\alpha}_k\}$, the proof follows.

The lemma follows. □

The following result shows that when α is sufficiently large, the seller optimizes profits under the blind box by pricing it so that consumers stop exactly after obtaining the special item.

Lemma E2. *For any given $\beta \geq 0$, there exists a threshold $\tilde{\alpha}^H$ such that for all $\alpha > \tilde{\alpha}^H$, the optimal blind box parameters are given by (p^*, q^*), with $p^* = q^* v_s(q^*, v_0)$ and*

$$q^* = \arg \max_{q \in (0, \frac{1}{n+1}]} \left(v_s(q, v_0) - \frac{c}{q} \right) = \frac{1}{1 + \sqrt{\frac{n\alpha v_0}{c}}},$$

where $v_s(q, v_0) = (1 + (1 - \frac{qn}{1-q})\alpha)v_0$ with $v_0 \in \{v^L, v^H\}$. Moreover, the expected number of blind boxes sold is given by $m_b^* = \frac{1}{q^*}$ if $v_0 = v^L$, and $m_b^* = \omega \frac{1}{q^*}$ if $v_0 = v^H$.

Proof of Lemma E2. The key insight underlying the proof is that as the status parameter $\alpha \to \infty$, the value of the special item, $v_s(v_0)$, becomes the dominant driver of the seller's profit. Therefore, under the optimal blind box, the seller ensures that the consumer keeps purchasing until they obtain the special item.

In what follows, we establish lower bounds on the seller's profit under two benchmark policies, where $p_b = q v_s(q, v^H)$ and $p_b = q v_s(q, v^L)$, respectively. We then show that for any alternative pricing policy, its profit is strictly dominated by at least one of these benchmarks as $\alpha \to \infty$. We assume

for the rest of the proof that $c > 0$; we deal with the $c = 0$ case separately at the end. Without loss of generality, we also assume that $v^L < v^H$.

Benchmark Policies.

We first note that an ω-fraction of consumers have a base reward of $v_0 = v^H$ for the special item, whereas the remaining $1-\omega$ fraction of consumers have a base reward of $v_0 = v^L$. Based on this, we define two benchmark policies based on whether the seller targets consumers with a high base reward ($v_0 = v^H$) or a low base reward ($v_0 = v^L$) for the special item.

1. **High-Price Benchmark**: Suppose that the blind box is priced at $p_b = q_H v_s(q_H, v^H)$ with $q_H = 1/(1+\sqrt{n\alpha v^H/c})$. Then, as $\alpha \to \infty$, the following inequality holds:

$$p_b = q_H v_s(q_H, v^H) = q_H \left(1 + \left(1 - \frac{q_H n}{1-q_H}\right)\alpha\right) v^H > (1-q_H)v^H + q_H v_s(q_H, v^L).$$

   Therefore, under this price, consumers whose base reward for the special item is $v_0 = v^L$ do not purchase at all, as the price exceeds their maximum possible valuation for the blind box. Moreover, since $p_b > (1-q_H)v^H$, a consumer who has already obtained the special item will cease purchasing. The expected number of purchases is then exactly ω/q_H. This gives us the following expression for the seller's profit under this policy.

$$\begin{aligned}\Pi_B^{(H)} &= \omega(v_s(q_H, v^H) - \frac{c}{q_H}) \\ &= \omega((1+\alpha)v^H - c - 2\sqrt{n\alpha v^H c}).\end{aligned} \tag{E.1}$$

2. **Low-Price Benchmark**: Alternatively, suppose that $p_b = q_L v_s(q_L, v^L)$ for $q_L = 1/(1+\sqrt{n\alpha v^L/c})$. Then, as $\alpha \to \infty$, we have $p_b > (1-q_L)v^H$. Therefore, all consumers stop purchasing once they obtain the special item. One can then infer that the expected number of blind boxes purchased by all consumers is exactly $1/q_L$. Based on this, we present a similar expression for the seller's profit under this policy, namely,

$$\begin{aligned}\Pi_B^{(L)} &= v_s(q_L, v^L) - \frac{c}{q_L} \\ &= (1+\alpha)v^L - c - 2\sqrt{n\alpha v^L c}.\end{aligned} \tag{E.2}$$

We divide the rest of the proof into three cases depending on the asymptotic behavior of q. Specifically, we consider the following three families of policies. We denote by Π'_B the profit under any alternative policy, and show that the benchmark policies generate strictly higher profits than Π'_B in each scenario: (i) policies (p_b, q) where $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = 0$, (ii) (p_b, q) where $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = z$ for some constant $z > 0$, and (iii) (p_b, q) where $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = \infty$.

Scenario (i): $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = 0$**.**

Suppose that we restrict our attention to the family of policies (p_b, q) satisfying $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = 0$. In this scenario, we have $\lim_{\alpha\to\infty} qv_s(q, v^H) = \lim_{\alpha\to\infty} q(1+(1-\frac{nq}{1-q})\alpha)v^H \to 0$ and therefore, any price $p_b \le qv_s(q, v^H)$ would be infeasible for sufficiently large α as $p_b < c$. Focusing on policies where $p_b > qv_s(q, v^H)$, the maximum profit under any such policy satisfies[13]

$$\Pi'_B \le (1+n\log n)(v^H - c),$$

which does not depend on α and is therefore dominated by the benchmark profit when α is sufficiently large.

Scenario (ii): $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = z$**.**

We analyze this scenario in three cases depending on the price p_b, namely: (a) $p_b > qv_s(q, v^H)$, (b) $qv_s(q, v^L) < p_b < qv_s(q, v^H)$, and (c) $p_b < qv_s(q, v^L)$.

Case (a): $p_b > qv_s(q, v^H)$**.** In this case, a sufficient condition for all consumers to stop purchasing is that they obtain all n regular items. Moreover, the maximum possible price to ensure non-zero purchases is $p_b = qv_s(q, v^H) + (1-q)v^H$. The profit can then be written as

$$\Pi'_B \le \frac{n(1+\log n)}{1-q}(qv_s(q, v^H) + (1-q)v^H - c).$$

Taking the limit as $\alpha \to \infty$, we get

$$\lim_{\alpha\to\infty} \Pi'_B \le (1+n\log n)(zv^H + v^H - c),$$

which is again independent of α and hence dominated by both the benchmark policies whose profits scale with α.

Case (b): $qv_s(q, v^L) < p_b < qv_s(q, v^H)$**.** Since $p_b > qv_s(q, v^L)$, we note that (the $1-\omega$ fraction of) consumers with a low base reward for the special item stop purchasing after obtaining all n unique regular items, and then they purchase blind boxes at most $\frac{n(1+\log n)}{1-q}$ times. On the other hand, consumers with a high base reward for the special item purchase at most $\frac{n(1+\log n)}{1-q} + \frac{1}{q}$ blind boxes, which is an upper bound on the total number of purchases required before all unique items are obtained. The profit then becomes

$$\Pi'_B \le \left(\frac{n(1+\log n)}{1-q} + \omega\frac{1}{q}\right)(qv_s(q, v^H) - c),$$

[13] We note that the expected number of blind boxes purchased by the consumer before obtaining all n unique items is $\frac{nH(n)}{1-q} \le \frac{n(1+\log n)}{1-q}$, where $H(n)$ denotes the n-th harmonic number, as is known from the Coupon Collector problem (Erdős and Rényi 1961).

Comparing to the benchmark policy and observing that $\lim_{\alpha\to\infty} qv_s(q,v^H) \to zv^H$, we get

$$\begin{aligned}
&\lim_{\alpha\to\infty} \frac{\Pi_B^{(H)}(\alpha) - \Pi'_B(\alpha)}{\alpha} \\
\geq &\omega v^H - \lim_{\alpha\to\infty} \frac{(\frac{n(1+\log n)}{1-q} + \omega\frac{1}{q})(qv_s(q,v^H) - c)}{\alpha} \\
= &\omega v^H - \lim_{\alpha\to\infty} \frac{qv_s(q,v^H) - c}{q\alpha} \\
= &\omega v^H - (\omega v^H - \omega\frac{c}{z}) \\
= &\omega\frac{c}{z} > 0,
\end{aligned}$$

which implies that $\Pi_B^{(H)}$ strictly dominates Π'_B when α is sufficiently large.

Case (c): $p_b < qv_s(q,v^L)$**.** In this case, we use $\frac{n(1+\log n)}{1-q} + \frac{1}{q}$ as an upper bound for the expected number of blind boxes purchased by all consumers, which gives us

$$\Pi'_B \leq (\frac{n(1+\log n)}{1-q} + \frac{1}{q})(qv_s(q,v^L) - c).$$

Comparing to the benchmark policy in the limit as $\alpha \to \infty$ leads to the following bound:

$$\begin{aligned}
&\lim_{\alpha\to\infty} \frac{\Pi_B^{(L)}(\alpha) - \Pi'_B(\alpha)}{\alpha} \\
\geq &\lim_{\alpha\to\infty} v^L - \frac{(\frac{n(1+\log n)}{1-q} + \frac{1}{q})(qv_s(q,v^L) - c)}{\alpha} \\
= &v^L - \lim_{\alpha\to\infty} \frac{qv_s(q,v^L) - c}{q\alpha} \\
= &v^L - (v^L - \frac{c}{z}) > 0.
\end{aligned}$$

where the final equality holds since $\lim_{\alpha\to\infty} qv_s(q,v^L) \to zv^L$, and $\lim_{\alpha\to\infty} q\cdot\alpha = z$. This completes our argument that the benchmark policy is more profitable.

Scenario (iii): $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = \infty$**.** This is arguably the most involved of the three scenarios, and we divide this into five cases based on the exact price of the blind box. Before diving into the cases, we note that for any q which satisfies $\lim_{\alpha\to\infty} q(\alpha)\cdot\alpha = \infty$, when α is sufficiently large, it must be the case that

$$qv_s(q,v^H) > qv_s(q,v^L) + (1-q)v^H \quad \text{and} \quad qv_s(q,v^L) > (1-q)v^H,$$

where $v_s(q,v_0) = (1 + (1 - \frac{nq}{1-q})\alpha)v_0$.

Case 1: $p_b > qv_s(q, v^H)$**.** In this regime, the expected number of blind boxes purchased satisfies $m \leq \min\left(\frac{n(1+\log n)}{(1-q)}, \frac{1}{q}\right)$ for consumers whose base reward for the special item is $v_0 = v^H$. Moreover, consumers with a low base reward for the special item do not purchase the blind box at all as $p_b > (1-q)v^H + qv_s(q, v^L)$. This allows us to state the following upper bound on the profit when $p_b > qv_s(q, v^H)$:

$$\begin{aligned}\Pi'_B(\alpha) &\leq \omega \min\left(\frac{n(1+\log n)}{1-q}, \frac{1}{q}\right)\left(qv_s(q, v^H) + (1-q)v^H - c\right)\\ &\leq \omega \min\left(n(1+\log n), \frac{1}{r}\right)\left(rv_s(q, v^H) + v^H - c\right),\end{aligned}$$

where $r = \frac{q}{1-q}$. If $\frac{1}{r} < n(1+\log n)$, we have

$$\Pi'_B(\alpha) \leq \omega\left((1+(1-nr)\,\alpha)v^H + \frac{v^H - c}{r}\right).$$

This bound is maximized when r takes on its smallest value, i.e., $r = \frac{1}{n(1+\log n)}$. Substituting this and taking the limit, we conclude that

$$\lim_{\alpha\to\infty} \frac{\Pi_B^{(H)}(\alpha) - \Pi'_B(\alpha)}{\alpha} \geq \omega v^H - (1 - \frac{1}{(1+\log n)})\omega v^H = \frac{\omega v^H}{1+\log n}.$$

Thus, the benchmark policy is more profitable. Next, suppose that $r < \frac{1}{n(1+\log n)}$. In this case, we have

$$\begin{aligned}\Pi'_B(\alpha) &\leq \omega n(1+\log n)\left(v^H + rv_s(q, v^H) - c\right)\\ &= \omega n(1+\log n)\left(v^H + rv^H(1+\alpha) - v^H\alpha r^2 n - c\right).\end{aligned}$$

The maximizer of the term $v^H + rv^H(1+\alpha) - v^H\alpha r^2 n - c$ is $\frac{1+\alpha}{2n\alpha}$ if $\frac{1+\alpha}{2n\alpha} \leq \frac{1}{n(1+\log n)}$; otherwise the maximum value is attained at the boundary $\frac{1}{n(1+\log n)}$.

Suppose that $\frac{1+\alpha}{2n\alpha} > \frac{1}{n(1+\log n)}$; taking the limit as $\alpha \to \infty$, we have

$$\lim_{\alpha\to\infty} \frac{\Pi_B^{(H)}(\alpha) - \Pi'_B(\alpha)}{\alpha} \geq \omega v^H - (1 - \frac{1}{(1+\log n)})\omega v^H = \frac{\omega v^H}{1+\log n}.$$

If $\frac{1+\alpha}{2n\alpha} \leq \frac{1}{n(1+\log n)}$, we have $n = 1$. Taking the limit as $\alpha \to \infty$, we obtain

$$\lim_{\alpha\to\infty} \frac{\Pi_B^{(H)}(\alpha) - \Pi'_B(\alpha)}{\alpha} \geq \omega v^H - \frac{1}{2}\omega v^H = \omega v^H \frac{3}{4} > 0.$$

Case 2: $p_b \in ((1-q)v^H + qv_s(q, v^L), qv_s(q, v^H))$**.** In this range, consumer behavior is identical to that in the high-price benchmark case discussed earlier. Specifically, only consumers with base reward $v_0 = v^H$ for the special item purchase the blind box, and they stop purchasing once they obtain the special item. For any parameters (p_b, q) in this range, the seller can strictly increase profits by raising the price to $p_b = qv_s(q, v^H)$ while keeping the same q. Hence, any such policy is suboptimal.

Case 3: $qv_s(q,v^L) < p_b \le (1-q)v^H + qv_s(q,v^L)$**.** In order to obtain an upper bound on the seller's profit in this case, we first characterize the expected number of blind boxes purchased by different consumer types. Specifically:

- Consumers whose base reward for the special item is given by $v_0 = v^H$ stop purchasing the blind box after they receive the special item, as $p_b > (1-q)v^H$. Their expected purchase quantity is at most $\frac{1}{q}$.
- For consumers with a low base reward for the special item ($v_0 = v^L$), they will stop purchasing if they receive the special item or (at most) all unique regular items. Their expected number of blind boxes purchased is at most $\min\{\frac{1}{q}, \frac{n(1+\log n)}{1-q}\}$.

The seller's profit under this pricing scenario, Π'_B, is bounded by

$$\begin{aligned}\Pi'_B(\alpha) &\le \left(\frac{\omega}{q} + (1-\omega)\min\left(\frac{1}{q}, \frac{n(1+\log n)}{1-q}\right)\right)\left((1-q)v^H + qv_s(q,v^L) - c\right)\\ &\le \left(\frac{\omega}{r} + (1-\omega)\min\left(\frac{1}{r}, n(1+\log n)\right)\right)\left(v^H + rv_s(q,v^L) - c\right),\end{aligned}$$

where $r = \frac{q}{1-q}$. First, suppose that $\frac{1}{r} \le n(1+\log n)$. Substituting this into the inequality above yields

$$\begin{aligned}\Pi'_B(\alpha) &\le \frac{1}{r}\cdot\left(v^H + rv_s(q,v^L) - c\right)\\ &= \left(\frac{v^H - c}{r} + v_s(q,v^L)\right).\end{aligned}$$

This expression is maximized when r takes its smallest value, namely $r = \frac{1}{n(1+\log n)}$. Substituting this, we conclude that

$$\Pi'_B(\alpha) \le n(1+\log n)(v^H - c) + v^L\left(1 + \alpha\left(1 - \frac{1}{1+\log n}\right)\right).$$

Therefore, we have

$$\lim_{\alpha\to\infty} \frac{\Pi_B^{(L)}(\alpha) - \Pi'_B(\alpha)}{\alpha} \ge v^L - v^L\left(1 - \frac{1}{1+\log n}\right) = \frac{v^L}{1+\log n} > 0,$$

which completes our argument that the benchmark policy is more profitable.

Next, suppose that $\frac{1}{r} > n(1+\log n)$. In this case, we have

$$\begin{aligned}\Pi'_B(\alpha) &\le \left(\frac{\omega}{r} + (1-\omega)n(1+\log n)\right)\left(v^H + rv_s(q,v^L) - c\right)\\ &\le \left(\frac{\omega}{r} + (1-\omega)n(1+\log n)\right)\left(v^H + rv^L(1+\alpha) - v^L\alpha r^2 n - c\right)\\ &\le U(r) + \omega v^L(1+\alpha) + (1-\omega)n(1+\log n)(v^H - c). \qquad \text{(E.3)}\end{aligned}$$

In the final expression, $U(r) = \omega\frac{v^H - c}{r} + (1-\omega)n(1+\log n)rv^L(1+\alpha) - \omega v^L \alpha r n$, and the final inequality holds since $-(1-\omega)v^L\alpha r^2 n^2(1+\log n)v^L < 0$. Our goal is to prove that for any given $r \in (0, \frac{1}{n(1+\log n)})$, either $\Pi_B^{(L)}(\alpha) - \Pi'_B(\alpha)$ or $\Pi_B^{(H)}(\alpha) - \Pi'_B(\alpha)$ is strictly positive as $\alpha \to \infty$.

Towards this goal, consider the expression $U(r)$ as defined earlier. Since its second derivative with respect to r is strictly positive, we have $U(r)$ is maximized at its extremes. When $r = \frac{1}{n(1+\log n)}$, we have $U(r = \frac{1}{n(1+\log n)}) = \omega n(1+\log n)(v^H - c) + (1-\omega)v^L(1+\alpha) - \omega\alpha\frac{v^L}{1+\log n}$. Substituting this back into Equation (E.3) and taking the limit as $\alpha \to \infty$, we have

$$\begin{aligned}\lim_{\alpha\to\infty} \frac{\Pi_B^{(L)}(\alpha) - \Pi'_B(\alpha)}{\alpha} &\geq v^L - \lim_{\alpha\to\infty} \frac{1}{\alpha}\left(v^L(1+\alpha) - \frac{1}{1+\log n}\omega v^L \alpha\right) \\ &= \omega\frac{v^L}{1+\log n}.\end{aligned}$$

Therefore, as long as $\omega > 0$, we can infer that $\Pi_B^{(L)}$ is strictly more profitable than Π'_B as $\alpha \to \infty$. When $\omega = 0$, the proof follows almost trivially as in the $\frac{1}{r} > n(1+\log n)$ case, so we do not repeat that here.

Next, consider $U(r)$ evaluated at $r \to 0^+$. First, we have

$$\lim_{\alpha\to\infty} \frac{U(r)}{\alpha} = \lim_{\alpha\to\infty} \omega\frac{v^H - c}{r\alpha} = 0.$$

In the final equation, we use the fact that $\lim_{\alpha\to\infty} \alpha r = \lim_{\alpha\to\infty} \frac{\alpha q}{1-q} \to \infty$. Taking the limit once again (against the high price benchmark), we get

$$\lim_{\alpha\to\infty} \frac{\Pi_B^{(H)}(\alpha) - \Pi'_B(\alpha)}{\alpha} \geq \omega v^H - \omega v^L > 0.$$

Therefore, we conclude that the benchmark policy dominates as $\alpha \to \infty$.

Case 4: $(1-q)v^H < p_b \leq qv_s(q, v^L)$**.** All consumers continue purchasing until they obtain the special item. This case is analogous to Case 1 as the seller can always increase their price to $p_b = qv_s(q, v^L)$ and strictly improve profits.

Case 5: $p_b \leq (1-q)v^H$**.** In this case, the price is sufficiently low that consumers may continue purchasing even after obtaining the special item. Based on our previous analysis, the seller's profit is bounded by

$$\begin{aligned}\Pi'_B &\leq \left(\frac{1}{q} + \frac{n(1+\log n)}{1-q}\right)\left((1-q)v^H - c\right), \\ &\leq \left(\frac{1}{r} + n(1+\log n)\right)\left(v^H - c\right),\end{aligned}$$

where $r = \frac{q}{1-q}$. We first evaluate the limit of the profit under this policy below.

$$\lim_{\alpha\to\infty} \frac{\Pi'_B(\alpha)}{\alpha} \le \lim_{\alpha\to\infty} \left[\left(v^H - c\right) \left(\frac{1}{\alpha r} + \frac{n(1+\log n)}{\alpha} \right) \right] = 0.$$

As a result, the benchmark policy strictly dominates any policy under case 5 as $\lim_{\alpha\to\infty} \frac{\Pi_B^{(L)}(\alpha)}{\alpha} = v^L$.

This completes the proof of Scenario (iii). In summary, as $\alpha \to \infty$, for any given (p_b, q) where $p_b \neq qv_s(v^L)$ and $p_b \neq qv_s(v^H)$, the given policy is suboptimal for all $c > 0$.

Finally, when $c = 0$, our argument above may not hold under scenario (ii). If $\lim_{\alpha\to\infty} q\alpha \to z$, for the profit to scale with α, it is therefore necessary for the expected number of purchases to increase (linearly) with α, which occurs only when $p_b \le qv_s(q, v^H)$. When $p_b < qv_s(q, v^L)$, the total profit can be written as follows:

$$\Pi'_B \le \left(\frac{1}{q} + n(1+\log n) \right) p_b.$$

We break this down into two cases. First, consider policies where $\lim_{\alpha\to\infty} p_b \le qv_s(q, v^L) - \epsilon$ for a constant ϵ that is independent of α. Then, it is easy to see that $\lim_{\alpha\to\infty} \frac{\Pi'_B}{\alpha} = v^L - \frac{\epsilon}{z}$, which is strictly dominated by $\Pi_B^{(L)}$. Second, when $\lim_{\alpha\to\infty} p_b = qv_s(q, v^L)$, we note that when α is sufficiently large, the stopping condition under p_b matches that under $p'_b = qv_s(q, v^L)$. As a result, the seller can simply increase the price to $p'_b = qv_s(q, v^L)$ and strictly improve profits. Thus, when α is sufficiently large, the profit under any policy where $p_b < qv_s(q, v^L)$ is strictly dominated by one of the benchmark policies. One can follow a similar rationale to show that any price $qv_s(q, v^L) < p_b < qv_s(q, v^H)$ is also strictly dominated by the high-value benchmark. This completes the proof. □

E.2. Other Auxiliary Results

Lemma E3. *Consider a special case of the base model with n regular items and no special item. Then the optimal profit under the blind box in the limit as the number of regular items approaches infinity is given by*

$$\lim_{n\to\infty} \frac{\Pi_B^*(n)}{n} \le \begin{cases} \frac{1}{e}\left(\omega(v^H - c) + (1-\omega)(v^L - c)\right) & \text{if } 0 \le c \le v^L, \\ \frac{\omega(v^H - c)}{e} & \text{if } v^L < c \le v^H. \end{cases}$$

Proof of Lemma E3. The proof builds on the results in Chen et al. (2021), which provides an exact characterization of the optimal profit under blind box (with no special item) in the costless regime. Specifically, Chen et al. (2021) show that as $n \to \infty$ and $c = 0$, the optimal profit in the absence of the special item becomes (see Theorem 4 in Chen et al. (2021))

$$\lim_{n\to\infty} \frac{\Pi_B^*(n, c=0)}{n} = \lim_{n\to\infty} \frac{(1-\omega)v^L + \omega v^H}{e} (H(n) - H(n/e)) = \frac{(1-\omega)v^L + \omega v^H}{e}, \tag{E.4}$$

where $H(n)$ is the n-th harmonic number. In order to extend this bound for a given $c \in [0, v^H]$, we construct an alternative instance where the item is costless, whose profit dominates that of the original instance. This allows us to construct the upper bound from the lemma statement. We divide the rest of the proof into two cases depending on whether $c \leq v^L$ or $c \in (v^L, v^H]$.

Case I: Low Cost ($0 \leq c \leq v^L$).

Consider an alternative instance I_1 where the consumer's valuation is $v^L - c$ with probability $1-\omega$ and $v^H - c$ with probability ω. Suppose that the cost is zero for this instance and let $\mathbb{E}[v'] = (1-\omega)(v^L - c) + \omega(v^H - c)$. Drawing on (E.4), the optimal profit under blind box for this instance satisfies: $\lim_{n\to\infty} \Pi_B^1/n = \mathbb{E}[v']/e$. We now show that this term serves as an upper bound for the optimal blind box profit under the original instance with cost c.

In order to prove this, we show that for any price p_b, the expected number of blind boxes purchased by consumers under the original instance with cost c is no greater than the number of blind boxes purchased by consumers in the alternative instance when the price of the blind box is $p_b' = p_b - c$. In the absence of the special item, let $0 \leq \ell_r \leq n$ denote the number of regular items that the consumer values at $v = v^H$ (i.e., the consumer type). Consider a consumer of type ℓ_r who has collected n^H high-value and n^L low-value items. Under the original instance, the consumer's expected value upon purchasing the next blind box is given by $E[V(n^L, n^H)] = \frac{\ell_r - n^H}{n} v^H + \frac{(n-\ell_r)-n^L}{n} v^L$. The consumer continues to purchase the blind box under the original instance as long as $E[V(n^L, n^H)] \geq p_b$.

On the other hand, consider a consumer of type ℓ_r under the alternative instance I_1 at the same state (n^L, n^H) faced with a blind box priced at $p_b' = p_b - c$. The consumer purchases the blind box as long as:

$$\frac{\ell_r - n^H}{n}(v^H - c) + \frac{(n-\ell_r) - n^L}{n}(v^L - c) \geq p_b - c.$$

The above inequality can be rewritten as $E[V(n^L, n^H)] - c\left(\frac{n-(n^H+n^L)}{n}\right) \geq p_b - c$, which simplifies to $E[V(n^L, n^H)] \geq p_b - c\left(\frac{n^H+n^L}{n}\right)$. Since $c \geq 0$, we infer the following: for any state (n^H, n^L), if a consumer purchases the blind box in the original instance at price p_b, then they must also purchase it in instance I_1 at price p_b'.

Since this claim holds for every consumer type ℓ_r and state (n^L, n^H), we conclude that the expected number of blind boxes purchased in the original instance at price p_b, $m(p_b)$, is no greater than in instance I_1 at price $p_b' = p_b - c$, i.e., $m_1(p_b - c)$. This allows us to bound the profit as follows:

$$\Pi_B(p_b) = m(p_b)(p_b - c) \leq m_1(p_b - c)(p_b - c) = \Pi_B^1(p_b - c).$$

Since this holds for any p_b, it must hold for the optimal price.

Case II: High Cost ($v^L < c \le v^H$).

The proof proceeds similarly in this case under a different alternative instance. Specifically, consider instance I_2 with the item costless where a consumer's valuation for a given regular item is $v^H - c$ with probability ω and 0 with probability $1-\omega$. Applying (E.4), the optimal profit in the limit satisfies $\lim_{n\to\infty} \Pi_B^2/n = \omega(v^H - c)/e$. Similar to our previous case, we show that for any price p_b for the original instance, the profit under the alternative instance is higher for price $p_b' = p_b - c$.

Once again, for a consumer of type ℓ_r in state (n^H, n^L), the consumer purchases the blind box under the original instance if $\frac{\ell_r - n^H}{n} v^H + \frac{(n-\ell_r)-n^L}{n} v^L \ge p_b$. Since $v^L < c$, this implies that

$$\begin{aligned} p_b - c &\le \frac{\ell_r - n^H}{n}(v^H - c) + \frac{(n-\ell_r)-n^L}{n}(v^L - c) - c\left(1 - \frac{n - n^L - n^H}{n}\right) \\ &\le \frac{\ell_r - n^H}{n}(v^H - c) + \frac{(n-\ell_r)-n^L}{n}(v^L - c), \end{aligned}$$

which is exactly the condition under which the consumer purchases the blind box in instance I_2 at price $p_b' = p_b - c$. As in Case I, this implies that consumers purchase at least as many items in the alternative instance. Therefore, $\Pi_B^2 \ge \Pi_B$. Taking the normalized limit provides the bound that the profit under the blind box at cost $c > v^L$ is at most $\frac{\omega(v^H - c)}{e}$. □

Lemma E4. *Let $\{g_i\}_{i\in I}$ be a family of convex functions defined on $\mathcal{C}$, and let $g^{\sup}(c) = \sup_{i\in I} g_i(c)$ for all $c \in \mathcal{C}$. Then, $g^{\sup}$ is convex on $\mathcal{C}$. Moreover, if all g_i are non-increasing (non-decreasing) on $\mathcal{C}$, then $g^{\sup}$ is non-increasing (non-decreasing) on $\mathcal{C}$ as well.*

Proof of Lemma E4. We show the convexity and monotonicity of $g^{\sup}(c)$ on $\mathcal{C}$ separately.

Convexity. For any $c_1, c_2 \in \mathcal{C}$ and $\theta \in [0,1]$, we have:

$$\begin{aligned} g^{\sup}\big(\theta c_1 + (1-\theta)c_2\big) &= \sup_i g_i\big(\theta c_1 + (1-\theta)c_2\big) \\ &\le \sup_i \big[\theta g_i(c_1) + (1-\theta) g_i(c_2)\big] \\ &\le \theta \sup_i g_i(c_1) + (1-\theta)\sup_i g_i(c_2) \\ &= \theta g^{\sup}(c_1) + (1-\theta) g^{\sup}(c_2). \end{aligned}$$

Hence $g^{\sup}(c)$ is convex.

Monotonicity. Assume that every g_i is non-increasing on $\mathcal{C}$. Take any $c_1, c_2 \in \mathcal{C}$ such that $c_1 \le c_2$. By monotonicity, we have

$$g_i(c_1) \ge g_i(c_2), \qquad \forall i \in I.$$

Suppose that $j \in I$ denotes the index such that $\sup_i g_i(c_2) = g_j(c_2)$. Then, we have

$$g^{\sup}(c_1) = \sup_i g_i(c_1) \ge g_j(c_1) \ge g_j(c_2) = g^{\sup}(c_2).$$

Hence $g^{\sup}$ is also non-increasing on $\mathcal{C}$. A similar argument holds when each g_i is non-decreasing. □

Lemma E5. *Consider an instance with n regular items valued at v and no special item. The optimal price p_b has the following form: $p_b(q,k) = (k+1)v/n$, where $k \in \{0,1,\ldots,n-1\}$. Correspondingly, the expected purchase quantity can be expressed as:*

$$m(k) = (1/n + 1/(n-1) \cdots + 1/(k+2) + 1/(k+1)) = n \sum_{i=k+1}^{n} 1/i.$$

Proof of Lemma E5. First, we show the form of the optimal price.

Define a state $n_r \in \{0,1,\ldots,n-1\}$ where the consumer has collected n_r unique regular items. If there are no special items and the valuation of each regular item is v, the consumer's marginal value from purchasing the blind box given state n_r can be rewritten as

$$v_B(n_r) = \frac{(n-n_r)}{n} v.$$

The consumer stops purchasing the blind box if and only if $v_B(n_r) < p_b$. Suppose that the consumers' stopping condition is met upon acquiring at least $\underline{n}$ regular items. This implies that the price of the blind box must satisfy the following condition:

$$v_B(\underline{n}) < p_b \leq v_B(\underline{n}-1).$$

As a result, the optimal price must be $v_B(\underline{n}-1)$. To see why, consider any price $p_b = v_B(\underline{n}-1) - \epsilon$ for some sufficiently small $\epsilon > 0$ so that this price meets the condition above. We note that under this new price, the consumer's stopping condition remains unchanged, and therefore, the expected number of purchases will not increase. Consequently, the seller can strictly increase profits by increasing the price by ϵ (reverting to $p_b = v_B(\underline{n}-1)$). In other words, the price $p_b = v_B(\underline{n}-1)$ dominates any price p_b satisfying $v_B(\underline{n}) < p_b < v_B(\underline{n}-1)$.

Second, we show the expected purchase number for any given price $p_b(q,k) = (k+1)v/n$.

Define the quantity $m(n_r)$ as the expected number of blind boxes purchased by consumers before reaching their stopping condition, when they initially possess $n_r \in \{0,1,\ldots,n\}$ unique regular items. Furthermore, given price $p_b = (k+1)v/n$, it is clear that the consumer would stop purchasing blind box if and only if they have obtained $n-k$ unique regular items; this is because the marginal value from any subsequent purchase is exactly $\frac{k}{n}v$, which is strictly smaller than the price.

For each purchase, there are two possible outcomes: (i) drawing a new regular item, in which case the state transitions to n_r+1, (ii) drawing a repeated item, in which case the state remains unchanged at n_r. Hence, we can write the recursion for the expected purchase volume as follows:

$$m(n_r) = 1 + \frac{n-n_r}{n} m(n_r+1) + \frac{n_r}{n} m(n_r),$$

with the boundary condition $m(n_r) = 0$ if $n_r \geq n - k$. In the above expression, the fractions $(n - n_r)/n$ and n_r/n represent the probability of drawing a new regular and repeated item, respectively. Solving the above recursion, we see that

$$m(n_r) = \frac{n}{n - n_r} + m(n_r + 1).$$

Using the fact that $p_b = \frac{k+1}{n} v$ and working backwards from $n_r = k$, we get

$$m(0) = \frac{n}{n} + \frac{n}{n-1} + \cdots + \frac{n}{k+2} + \frac{n}{k+1} = n \sum_{i=k+1}^{n} \frac{1}{i}.$$

Note that $m(0)$ denotes the expected number of purchases a consumer will make from an initial state with no items. □

Lemma E6. *Suppose that $g(k) = (k+1)(H_n - H_k)$ for $k \in \{0, 1, \ldots, n-1, n\}$, where H_k is the k-th harmonic number with $H_0 = 0$ by definition. Then, for every integer $n \geq 1$ and $k \in \{0, 1, \ldots, n-1, n\}$, we have*

$$g(k) \leq \frac{n}{e} + 3.$$

Proof of Lemma E6. First, we note that for all n, $g(0) = H_n$, which is bounded from above by $\frac{n}{e} + 3$. Furthermore, for all $k \geq 1$, we use the fact that

$$g(k) = (k+1)(H_n - H_k) \leq (k+1)(\log(n) - \log(k)) = g_2(k).$$

We prove that $g_2(x) \leq \frac{n}{e} + 3$ in the continuous range $[1, n]$, which implies the lemma. Optimizing for this function $g_2(x)$ in the range $[1, n]$, we have

$$g_2'(x) = \log(n) - \log(x) - \frac{x+1}{x}, \tag{E.5}$$

and $g_2''(x) = \frac{1}{x}(\frac{1}{x} - 1) \leq 0$.

Suppose that x^* denotes the value of x at which the derivative evaluates to zero. Then, a slight rearrangement gives us

$$\log\left(\frac{n}{x^*}\right) = \frac{x^* + 1}{x^*}. \tag{E.6}$$

First, we see that $g_2'(1) = \log(n) - 2 \geq 0$ for all $n \geq e^2$. Similarly, $g_2'(\frac{n}{e}) = -\frac{e}{n} < 0$ for all $n \geq 1$. We consider two cases depending on the value of n:

1. If $n \leq \lfloor e^2 \rfloor = 7$, then $g_2(x)$ is decreasing in $x \in [1, n]$, and $g_2(x) \leq g_2(1) = 2H_n$. One can verify that $2H_n \leq \frac{n}{e} + 3$ for all n.

2. If $n \geq 8$, then we know that from Equation (E.5) that $x^* \leq \frac{n}{e}$. Further, from Equation (E.6), it must be the case that $g_2'(x^*) = 0$, which implies that $\log(n) - \log(x^*) = \frac{x^*+1}{x^*}$. Plugging this back into our expression for $g_2(x)$, we get

$$\begin{aligned} g_2(x^*) &= (x^*+1) \cdot \frac{x^*+1}{x^*} \\ &= x^* + 2 + \frac{1}{x^*} \\ &\leq \frac{n}{e} + 2 + \frac{1}{n/e} \\ &\leq \frac{n}{e} + 3. \end{aligned}$$

In the first inequality, we use the fact that $x^* \leq \frac{n}{e}$ and that the expression $x^* + 2 + \frac{1}{x^*}$ is increasing in the value of x^*.

This completes the proof of the lemma. □

Lemma E7. *Suppose that the lump-sum bonus for collecting all $n+1$ items is sufficiently large, i.e., $v^+ \geq (n+1+\alpha)v^H \frac{z}{z-1}$, where $z = \frac{H_{n+1}}{H_n}$. For any given $q \in \left(0, \frac{1}{n+1}\right]$, suppose that $p_b > \frac{nv^L + v_s(q, v^L) + v^+}{\bar{m}}$, where $\bar{m}$ denotes the expected number of purchases required to collect all $n+1$ items. Then, the fraction $\bar{f}_B$ of consumers that do not collect the bonus satisfies*

$$\bar{f}_B \geq (1-\omega)^{n+1} \cdot \frac{1}{n+1}.$$

Proof of Lemma E7. To prove the result, it suffices to consider the "worst" consumer type, who values every item at v^L (including the base reward of the special item) and occurs with probability $(1-\omega)^{n+1}$. Since all other types have weakly higher valuations, showing that this type fails to collect the bonus with probability at least $\frac{1}{n+1}$ immediately establishes the stated lower bound for all consumer types.

Recall that consumer types are denoted by (ℓ_r, ℓ_s), where ℓ_r is the number of regular items with valuation $v_i = v^H$ and $\ell_s = 1$ indicates $v_0 = v^H$. The worst type corresponds to $(\ell_r, \ell_s) = (0,0)$. Thus, it remains to show that, for type $(0,0)$, the probability of not collecting the bonus is at least $\frac{1}{n+1}$ for any $q \in \left(0, \frac{1}{n+1}\right]$ and any price

$$p_b > \frac{nv^L + v_s(q, v^L) + v^+}{\bar{m}}.$$

Here, $\bar{m}$ denotes the expected number of purchases required to collect all $n+1$ items and depends only on q. For convenience, define

$$V^L_{\text{total}} = nv^L + v^L\left(1 + \alpha\left(1 - \frac{qn}{1-q}\right)\right) + v^+.$$

We now establish the desired claim for any price of the form $p_b = \frac{V^L_{\text{total}}+\epsilon}{\bar{m}}$ with $\epsilon > 0$ sufficiently small. Since the lemma holds for prices that are arbitrarily close to $\frac{V^L_{\text{total}}}{\bar{m}}$, it must also hold for all higher prices $p_b > \frac{V^L_{\text{total}}}{\bar{m}}$ for a given q. The reason is that, as the price increases, the consumer's stopping time can only move earlier, and therefore the probability of obtaining the bonus is weakly decreasing for every consumer type.

Our main idea involves two key arguments. In the first argument, we show that under any price $p_b = \frac{V^L_{\text{total}}+\epsilon}{\bar{m}}$, consumers of type $(0,0)$ do not always receive the bonus. Specifically, if such consumers were to follow a policy of purchasing until completion, their expected utility would be negative. To see this, observe that the expected value these consumers obtain from collecting all $n+1$ items is $V^L_{\text{total}} = nv^L + v^L\left(1+\alpha\left(1-\frac{qn}{1-q}\right)\right) + v^+$. Since completing the collection requires $\bar{m}$ blind box purchases in expectation, their expected net utility is

$$V^L_{\text{total}} - \bar{m}\cdot p_b < V^L_{\text{total}} - \bar{m}\cdot\frac{V^L_{\text{total}}}{\bar{m}} = 0.$$

Consequently, consumers of type $(0,0)$ cannot achieve the bonus on every sample path. Naturally, they may still collect the bonus with some non-zero probability if they do not follow a policy of purchasing until completion. The second argument bounds this probability by proving that the consumer always stops on sample paths where they have not yet collected the special item.

Formally, let (n_r, n_s) denote the consumer's state along any sample path, where $n_r \in \{0,\dots,n\}$ is the number of unique regular items obtained and $n_s \in \{0,1\}$ indicates whether the special item has been collected. Let $U(n_r, n_s)$ be the optimal value function (expected future utility) at state (n_r, n_s) for a consumer of type $(0,0)$. We claim that such consumers will never stop at any state of the form $(k,1)$ with $k < n$, that is, after obtaining the special item but before collecting all regular items. To see this, note that if the consumer continues purchasing from state $(k,1)$ until all remaining items are collected, the total value obtained is exactly $(n-k)v^L + v^+$. Moreover, the expected number of purchases required to collect the remaining $n-k$ regular items is at most $\frac{nH_n}{1-q}$, where H_n denotes the n-th harmonic number. This bound follows directly from the classical coupon collector argument. Thus,

$$\begin{aligned}
U(k,1) &\geq (n-k)v^L + v^+ - \frac{V^L_{\text{total}}+\epsilon}{\bar{m}}\cdot nH_n\frac{1}{1-q} \\
&\geq (n-k)v^L + v^+ - \frac{V^L_{\text{total}}+\epsilon}{(n+1)H_{n+1}}\cdot nH_n\frac{n+1}{n} \\
&\geq (n-k)v^L + v^+ - \frac{nv^L + v^L(1+\alpha) + v^+}{z} \\
&= (n-k)v^L + v^+(1-\frac{1}{z}) - \frac{nv^L + v^L(1+\alpha)}{z}
\end{aligned}$$

$$\geq (n-k)v^L + nv^H + v^H(1+\alpha) - \frac{nv^L + v^L(1+\alpha)}{z}$$
$$> 0.$$

In the second inequality, we use the fact that $m \geq (n+1)H_{n+1}$, a lower bound on the number of blind boxes obtained when $q = \frac{1}{n+1}$, and $1/(1-q) \leq (n+1)/n$. Recall that $z = \frac{H_{n+1}}{H_n}$, which is a constant for any fixed n. In the penultimate inequality, we use the fact that $v^+ \geq v^H(n+1+\alpha)\frac{z}{z-1}$. In conclusion, we have proved that $U(k,1) > 0$ for all $k \in \{0,1,2,\ldots,n-1\}$ and therefore, consumers will never stop purchasing at any state of the form $(k,1)$.

Thus far, we have established two facts: (i) consumers of type $(0,0)$ will stop before collecting all $n+1$ items, and (ii) they will never stop at any state $(k,1)$ in which the special item has already been obtained. Consequently, such a consumer must stop at some state of the form $(k,0)$ with $k \leq n-1$, i.e., before the special item is collected. To complete the argument, it remains to show that the probability of reaching a state $(k,0)$ for any $k \leq n-1$ is at least $\frac{1}{n+1}$.

Formally, let $k_{\min} \leq n-1$ denote the smallest value of k such that $(k,0)$ denotes a stopping state. Given that the consumer always continues purchasing at states $(k,1)$, the event that the bonus is not collected is precisely the event that their sample path reaches state $(k_{\min},0)$ (note that once the consumer collects the special item, they cannot reach state $(k_{\min},0)$). We now establish a lower bound on the probability of the consumer reaching such a state. Let $\Pr(E_{k,0} \mid p'_b, q')$ denote the probability that the consumer reaches state $(k,0)$ under blind box parameters (p'_b, q'). As a detour, consider an alternative instance with $p'_b = 0$ and $q' = \frac{1}{n+1}$, in which all $n+1$ items are equally likely and the consumer continues purchasing blind boxes until all $n+1$ items have been collected. In this benchmark instance, we have $\Pr(E_{n,0}) = \frac{1}{n+1}$, because the probability that the special item is drawn last is the same as the probability that any other item is drawn last. We can now use this benchmark to bound our desired probability as follows:

$$\Pr(E_{k_{\min},0}|p_b,q) = \Pr(E_{k_{\min},0}|0,q) \geq \Pr(E_{n,0}|0,q) \geq \Pr(E_{n,0}|0,\frac{1}{n+1}) = \frac{1}{n+1}.$$

We explain each step carefully. The equality follows because, under the optimal policy for the given blind box (p_b,q), the consumer never stops before reaching $(k_{\min},0)$, and therefore, the purchase behavior would not diverge prior to that state when compared to the instance with price $p'_b = 0$. Moreover, if we hold price and probability q fixed, then the probability that the consumer reaches state $(k,0)$ is weakly decreasing in k, which explains the second inequality. The final inequality is crucial: making the special item rarer (i.e., decreasing q') can only increase the probability that all n regular items are collected before the special item. In conclusion, we have

under blind box parameters (p_b, q) with $p_b = \frac{V^L_{\text{total}}+\epsilon}{\bar{m}}$, the probability that the consumer reaches the stopping state $(k_{\min}, 0)$ is at least $\frac{1}{n+1}$.

We can now complete the proof. From the previous steps, the probability that a consumer of type $(0, 0)$ does not collect the bonus is at least $\frac{1}{n+1}$. Since this consumer type occurs with probability $(1-\omega)^{n+1}$, the lemma follows. $\square$